

Beyond Nodal Damping: Collective Interaction Bounds for Delay-Aware SG–GFL Co-Design

Jorge Luis Mayorga Taborda
Universidad de los Andes, Colombia

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Abstract

Local PLL tuning and measurement delay act through the whole electrical network. Starting from the exact delayed linearization, we show that changing one PLL delay produces a rank-one update of the physical torque–speed impedance. With independent propagation paths and a nonzero update, its Hermitian damping increment has one positive and one negative eigenvalue: latency redistributes harmonic damping across collective directions. For a fixed-equilibrium SG/GFL sharing contract, retaining network algebraic variables also gives an affine descriptor perturbation. A contour identity yields finite spectral-cluster changes, their full parameter Hessian, and an explicit closed-walk remainder. Necessary cluster conditions lead to linear and semidefinite outer relaxations of secure replacement, conditional on verified contour and remainder bounds. In IEEE–39, all 60 declared harmonic tests confirm the finite-delay identity numerically. Three frozen algebraic points give walk remainder bounds 18–46 times smaller than a scalar-norm bound. Complex ball arithmetic certifies one root cluster throughout a declared parameter box. A preserved analytical gain construction at 40 ms delay yields 87.6% GFL and passes five nonlinear events. A new matched Julia comparison finds that unchanged gains also pass all five, with small opposing frequency and RoCoF changes. The results establish conditional interaction laws and a feasible construction; they do not establish replacement superiority or a physically limiting maximum.

Index terms: grid-following converters, PLL delay, descriptor systems, nonlinear eigenvalues, collective interactions, regional replacement bounds.

1 Introduction and research question

Replacing synchronous generation changes inertia, electrical coupling and converter synchronization simultaneously. Retuning each PLL against one operating point can preserve an assigned oscillatory mode while moving other modes toward a security boundary. The design question is therefore quantitative: within a declared replacement and gain region, how much replacement can any admissible local PLL tuning support, at fixed physical delays and under a stated security contract?

Detailed low-inertia models already connect converter control settings to collective modal behavior [17]. Grid structure and PLL synchronization are also established concerns [4]. Stability-constrained planning incorporates frequency and grid-strength constraints [18]; recent expansion-planning work explicitly addresses converter-driven modes [19]. Thus neither PLL tuning, delay-induced instability, nor the use of network information is by itself a new contribution.

The methodological distinction examined here is a *necessary finite regional bound across all allowed local gains*, rather than only a successful tuning or a fixed-pattern assignment obstruction. Descriptor lifting [20], nonlinear-eigenvalue contour localization [21], contour moments [14], structured DDE robustness [22], and fixed-structure delayed-controller optimization [16] are antecedents, not

discoveries of this work. The proposed synthesis specializes them to the actual SG/GFL sharing model and makes inter-site curvature and remainder terms explicit.

The manuscript establishes four conditional mathematical results: a finite physical damping-redistribution identity, an affine full-model intervention representation, a cluster curvature/closed-walk bound, and a controller-bounded necessary replacement relaxation. Its numerical evidence includes identity and enclosure tests of the bound machinery, a frozen full-model feasible retuning step, and a matched five-event fixed-gain follow-up. An informative IEEE-39 replacement ceiling below the parameter-box ceiling has not been demonstrated. This limitation prevents a claim of superior replacement capacity or certified near-optimality.

Organization. The main text states the physical contract, derives the finite regional bounds, and reports their evidence and limitations. Appendices preserve the physical damping derivation, frequency-moment law, nonlinear RoCoF argument, exact gain map, and local design constructions. These auxiliary results have their own assumptions; they are not combined into an unstated nonlinear stability theorem.

2 Physical model and optimization contract

With fixed device support, write the nonlinear retarded DAE as

$$E(x, \rho)\dot{x} = f_0(x, y; \rho) + \sum_i b_i(K_{p,i}, K_{i,i})e_i(t - \tau_i), \quad (1)$$

$$0 = g(x, y; \rho, d), \quad e_i = -\sin \theta_i v_{r,i} + \cos \theta_i v_{i,i}. \quad (2)$$

The exported reduced state convention has $E = I$. Physical inertia is recovered by consistent rescaling of SG swing rows, not by mixing row conventions. The implemented PLL is exactly

$$\dot{\theta}_i = \omega_{p,i}, \quad t_{f,i}\dot{\omega}_{p,i} = \xi_i + K_{p,i}e_i(t - \tau_i) - \omega_{p,i}, \quad \dot{\xi}_i = K_{i,i}e_i(t - \tau_i). \quad (3)$$

Here $t_{f,i} = 1/(300\,2\pi)$ s is the existing filter constant, distinct from pure delay τ_i . Other converter equations use θ_i but do not directly use $\omega_{p,i}$ or ξ_i in this implementation. No current-loop, setpoint or DC signal is delayed by this contract.

Use initialized generator dispatch and rating:

$$\epsilon_i = 1 - \rho_i, \quad P_{SG,i} = \epsilon_i P_i^0, \quad P_{GFL,i} = \rho_i P_i^0, \quad M_i = 2H_i S_i^0 \epsilon_i, \quad J_{SG} = \sum_i P_i^0 \epsilon_i.$$

S_i^0 and P_i^0 are different quantities. Replacement changes electrical injection/coupling as well as inertia. State removal at $\rho_i = 0$ or 1 requires another support chart.

For consistent per-unit phasors $P_i + iQ_i = V_i I_i^*$. The lossy network and voltage/current/DC dynamics remain inside g, f . If g_y is nonsingular, algebraic elimination gives

$$\delta y = -g_y^{-1}(g_x \delta x + g_d \delta d), \quad A_{\text{inst}} = f_x - f_y g_y^{-1} g_x$$

in ODE coordinates, with the analogous substitution in each detector. Thus P/Q and voltage effects enter the actual characteristic operator; a lossless Laplacian does not replace the AC Jacobian. No converter current limiter is implemented, so no limiter safety is claimed.

Sources inspected include:

- `experiments/nonlinear_codesign_20261001/ReducedDAE.jl`;

- `src/bnd_model_expN/PDExactDesignN.jl`;
- `experiments/graph_gsp_codesign_20261003/DelayedEvents.jl`.

The intended DC convention is physical supply in `ReducedDAE.jl`; the older helper by itself uses a different convention.

For a compact support-preserving region $\mathcal{D}$, define

$$P_{\mathcal{D}}^*(\tau) = \sup_{(\rho, K_p, K_i) \in \mathcal{D}} (P^0)^T \rho$$

over designs satisfying the complete required nonrotational spectral margin and every declared nonlinear-event constraint. The delays τ_i are exogenous. A finite event set is a specified security test, not robustness to arbitrary disturbances. The initialized dispatch $(P^0)^T \mathbf{1}$ is the denominator for the replacement fraction; net load and installed capacity are not substituted for it.

All identities involving fixed exported matrices apply to the repository's fixed-support, fixed-equilibrium current-sharing chart. The internal per-unit device equations and trims are held fixed while their terminal-current contributions are scaled. Physical rating/inertia interpretation must use that same normalization. Redispatch, topology changes, removal of device states at endpoints, or different nonlinear converter models require a new derivation or explicitly bounded model terms.

3 A finite regional replacement bound with moving modal patterns

A regional necessary bound must permit the modal patterns to move with the design. The following construction uses complete analytic characteristic matrices and bounds root clusters on fixed contours. It does not impose the eigenvector compatibility conditions used by the separate gain-construction method.

3.1 A finite identity that survives internal root collisions

Status: `EXACT_IDENTITY`. Use the full finite-dimensional characteristic matrix $\Delta(s, p)$ of the linearized DDE, with fixed delays. It is holomorphic in s on and inside each positively oriented, piecewise smooth simple contour Γ used below. Do not use a meromorphic Schur complement without accounting for its poles. Exclude the rotational root from these contours. In the full state chart this avoids requiring a parameter-independent gauge-deflation matrix.

Let $p = p_0 + d$ lie in a declared compact design region $\mathcal{D}$ and define $\mathcal{D}_d = \{d : p_0 + d \in \mathcal{D}\}$, and let $\Delta_0(s) = \Delta(s, p_0)$ be nonsingular on Γ . Suppose a factorization

$$\Delta(s, p) - \Delta_0(s) = \mathcal{U}(s)\mathcal{V}(s, d), \quad H(s, d) = \mathcal{V}(s, d)\Delta_0(s)^{-1}\mathcal{U}(s)$$

is available with $H \in \mathbb{C}^{r \times r}$, $\mathcal{V}(s, 0) = 0$. All dimensions and coordinates remain fixed on $\mathcal{D}$. Assume the *uniform* bound

$$\sup_{d \in \mathcal{D}_d, s \in \Gamma} \|H(s, d)\|_2 \leq \kappa_\Gamma < 1. \quad (4)$$

The determinant lemma and invertibility of $I + tH$, $0 \leq t \leq 1$, preserve the number $m_\Gamma > 0$ of enclosed roots, counting multiplicity. Define their mean $z_\Gamma(p) = m_\Gamma^{-1} \sum_{\lambda \text{ inside } \Gamma} \lambda$. Then

$$\boxed{z_\Gamma(p) - z_\Gamma(p_0) = -\frac{1}{2\pi i m_\Gamma} \oint_\Gamma \text{tr} \log(I + H(s, d)) ds.} \quad (5)$$

Proof. The determinant ratio on the contour is $\det(I+H)$. The norm-convergent series $\text{tr} \log(I+H) = \sum_{\ell \geq 1} (-1)^{\ell+1} \text{tr}(H^\ell)/\ell$ chooses a continuous, single-valued logarithm along the closed contour. The argument principle gives the difference of root sums as $(2\pi i)^{-1} \oint s d \log \det(I+H)$. Integration by parts gives (5). This only uses regularity on the contour and constant enclosed count; roots may collide or become defective inside it. $\square$

If every required nonrotational physical root satisfies $\Re \lambda \leq -\sigma$, then necessarily

$$g_\Gamma(p) := \Re z_\Gamma(p) + \sigma \leq 0.$$

The converse is false: a stable mean can hide an unstable member. Several disjoint contours can sharpen the necessary conditions. One-root contours recover individual sensitivities where regular. A finite collection is sufficient for a *necessary relaxation*, not for a full-spectrum feasibility claim.

3.2 An explicit uniform remainder

Status: PROVED_REDUCED_MODEL. Write $H = H_1 + H_{\text{rem}}$ with H_1 linear in d , and suppose

$$\sup_{\Gamma, \mathcal{D}_d} \|H_{\text{rem}}\|_* \leq e_\Gamma, \quad \|\cdot\|_* \text{ is the nuclear norm.}$$

Let ℓ_Γ be the contour length and put $\varphi(x) = -\log(1-x) - x$, $0 \leq x < 1$. The affine prediction and its remainder satisfy

$$g_\Gamma(p) = g_\Gamma(p_0) + J_\Gamma d + \mathcal{R}_\Gamma(d), \quad (6)$$

$$J_\Gamma d = -\Re \frac{1}{2\pi i m_\Gamma} \oint_\Gamma \text{tr} H_1(s, d) ds, \quad (7)$$

$$\boxed{|\mathcal{R}_\Gamma(d)| \leq \beta_\Gamma := \frac{\ell_\Gamma}{2\pi m_\Gamma} [e_\Gamma + r\varphi(\kappa_\Gamma)]}. \quad (8)$$

Proof. The trace of $H - H_1$ is bounded by its nuclear norm. Also $|\text{tr} H^\ell| \leq r \|H\|_2^\ell$, including nonnormal H . Summing terms $\ell \geq 2$ gives $r\varphi(\kappa_\Gamma)$; contour integration proves the bound. $\square$

A simpler conservative expression uses $\varphi(x) \leq x^2/[2(1-x)]$. For the logarithm term, any consistently used induced matrix norm can replace the Euclidean norm: each eigenvalue of H is bounded by that norm, hence $|\text{tr} H^\ell| \leq r\kappa^\ell$. Pointwise verified bounds may instead be integrated along Γ . Numerical errors in $g_\Gamma(p_0)$ and J_Γ must be included: if their certified errors are e_0 and $e_{J,a}$ and $|d_a| \leq r_a$, add $e_0 + \sum_a e_{J,a} r_a$ to β_Γ . No eigenvalue Hessian or fixed modal pattern is required.

3.3 The actual replacement contract gives a small factorization

Status: EXACT_IDENTITY. For the repository's exported fixed-support, fixed-equilibrium sharing chart, define

$$\begin{aligned} W_\rho &= \text{diag}(1 - \rho_1, 1 - \rho_1, \dots, 1 - \rho_n, 1 - \rho_n), \\ G(\rho) &= Y + W_\rho D_s, & N(\rho) &= W_\rho C_s + (I - W_\rho) C_f, \\ V(\rho) &= -G(\rho)^{-1} N(\rho), & C(\rho) &= E_\theta + H_v V(\rho), \\ B(K) &= B_p \text{diag} K_p + B_i \text{diag} K_i, & E_\tau(s) &= \text{diag}(e^{-s\tau_i}). \end{aligned}$$

Here V is the algebraic voltage *Jacobian*, not the voltage phasor. With ordinary gain coordinates,

$$\Delta(s, p) = sI - A_{\text{dev}} - B_v V(\rho) - B(K)E_\tau(s)C(\rho).$$

This is the inspected parametric model, not a universal identity for all SG/GFL models or dispatch changes. If equilibrium/device matrices change under another contract, their derivatives and remainders must also be bounded. Endpoint state removal requires a separate chart.

Denote reference matrices by a subscript 0, and set

$$J = G_0^{-1}\delta G, \quad V_1 = -G_0^{-1}(\delta N + \delta G V_0), \quad \delta V = (I + J)^{-1}V_1, \quad V_{\text{rem}} = -(I + J)^{-1}J V_1.$$

Thus $\delta V = V_1 + V_{\text{rem}}$ exactly. If uniformly $\|J\|_2 \leq a < 1$ and $\|V_1\|_2 \leq v$, then

$$\|\delta V\|_2 \leq \frac{v}{1-a}, \quad \|V_{\text{rem}}\|_2 \leq \frac{av}{1-a}. \quad (9)$$

Because G, N are affine in ρ , computable bounds are

$$a = \sum_j r_{\rho,j} \|G_0^{-1}G_{\rho_j}\|_2, \\ v = \sum_j r_{\rho,j} \|G_0^{-1}(N_{\rho_j} + G_{\rho_j}V_0)\|_2.$$

These require verified solves and norm enclosures for a certificate.

Set $D_0(s) = B_v + B_0 E_\tau(s) H_v$ and define

$$\mathcal{U}(s) = [D_0(s) \quad B_p E_\tau(s) \quad B_i E_\tau(s)], \quad (10)$$

$$\mathcal{V}(s, d) = - \begin{bmatrix} \delta V \\ \text{diag}(\delta K_p)(C_0 + H_v \delta V) \\ \text{diag}(\delta K_i)(C_0 + H_v \delta V) \end{bmatrix}. \quad (11)$$

Direct expansion proves $\Delta - \Delta_0 = \mathcal{U}\mathcal{V}$. The linear part uses V_1, C_0 in the three block rows. The remainder uses respectively $V_{\text{rem}}, \text{diag}(\delta K_p)H_v \delta V$, and $\text{diag}(\delta K_i)H_v \delta V$, with the same minus signs. For this export, $\mathcal{U}$ has 40 columns for 204 states: 20 voltage coordinates and two sets of ten PLL gain channels. The dimension is not a discarded-state approximation.

For explicit conservative constants, let $v_b = v/(1-a)$, $v_r = av/(1-a)$, $k_p = \max_i |\delta K_{p,i}|_{\text{max}}$, $k_i = \max_i |\delta K_{i,i}|_{\text{max}}$, and $c_b = \|C_0\|_2 + \|H_v\|_2 v_b$. Then

$$\|\mathcal{V}\|_2 \leq v_{\mathcal{V}} := \sqrt{v_b^2 + (k_p c_b)^2 + (k_i c_b)^2}, \\ \|\mathcal{V}_{\text{rem}}\|_2 \leq v_{\text{rem}} := \sqrt{v_r^2 + (k_p \|H_v\|_2 v_b)^2 + (k_i \|H_v\|_2 v_b)^2}.$$

If w_Γ uniformly bounds $\|\Delta_0(s)^{-1}\mathcal{U}(s)\|_2$ on the entire contour, use $\kappa_\Gamma = v_{\mathcal{V}} w_\Gamma$ and $e_\Gamma = r v_{\text{rem}} w_\Gamma$. Fixed nonsingular state/port scalings may improve these bounds if applied consistently. A large bound is inconclusive, not instability.

3.4 Affine descriptor realization with no parameter remainder

Status: EXACT_IDENTITY. The inverse-network remainder can be avoided altogether for this sharing contract. Retain the algebraic voltage perturbation v and the detector perturbation e with the dynamic state x :

$$\mathcal{F}(s, p) = \begin{bmatrix} sI - A_{\text{dev}} & -B_v & -B(K)E_\tau(s) \\ N(\rho) & G(\rho) & 0 \\ -E_\theta & -H_v & I \end{bmatrix}. \quad (12)$$

These are additional algebraic coordinates, not invented dynamics or additional control channels. Eliminating e and v proves

$$\det \mathcal{F}(s, p) = \det G(\rho) \det \Delta(s, p). \quad (13)$$

On a region where G is nonsingular, the finite roots and their multiplicities coincide. The factor $\det G$ is independent of s .

Define $d = (\delta\rho, \delta K_p, \delta K_i)$ and the repeated diagonal

$$\Theta(d) = \text{diag}(\delta\rho_1, \delta\rho_1, \dots, \delta\rho_n, \delta\rho_n, \delta K_p, \delta K_i).$$

With the reference operator $\mathcal{F}_0$, set

$$\mathcal{U} = \begin{bmatrix} 0 & B_p & B_i \\ I & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathcal{V}(s) = - \begin{bmatrix} C_s - C_f & D_s & 0 \\ 0 & 0 & E_\tau(s) \\ 0 & 0 & E_\tau(s) \end{bmatrix}, \quad (14)$$

$$T(s) = \mathcal{V}(s)\mathcal{F}_0(s)^{-1}\mathcal{U}, \quad H(s, d) = \Theta(d)T(s). \quad (15)$$

Then, exactly,

$$\mathcal{F}(s, p) - \mathcal{F}_0(s) = \mathcal{U}\Theta(d)\mathcal{V}(s), \quad \frac{\det \mathcal{F}(s, p)}{\det \mathcal{F}_0(s)} = \det(I + \Theta(d)T(s)).$$

The descriptor has dimension 234 and the determinant update has dimension 40 for this export. In particular,

$$\boxed{H = H_1, \quad H_{\text{rem}} = 0.}$$

All nonlinear dependence of the finite root change is now in the logarithm. The ordinary gain coordinates matter: log-gain increments would require exponential parameter remainders. Applying (5) to $\mathcal{F}$ gives the same physical root means, and (8) simplifies to

$$\boxed{\beta_\Gamma = \frac{\ell_\Gamma r}{2\pi m_\Gamma} \varphi(\kappa_\Gamma).} \quad (16)$$

The baseline-moment and coefficient integration errors still need their separate enclosures.

Use the declared parameter box, not a product of large norms. For increments $l_a \leq d_a \leq u_a$, put $r_a = \max(|l_a|, |u_a|)$. Let R_d contain these nonnegative displacement radii, repeated for the two occurrences of each replacement coordinate. They equal box half-widths only when p_0 is its center. Entrywise,

$$|H(s, d)| \leq M(s) := R_d |T(s)|.$$

For any fixed vector $w > 0$ on a contour panel,

$$\|H(s, d)\|_{w, \infty} \leq \max_a \frac{(M(s)w)_a}{w_a} =: \kappa_w(s). \quad (17)$$

This follows by scaling with $\text{diag } w$ and taking absolute row sums. It is a standard positive-matrix bound, not a new stability theorem. The repeated-coordinate correlation is relaxed conservatively; the controllers remain local and the physical gain bounds unchanged. A Perron-based weight can suggest w , but certification uses verified inequalities for the chosen positive vector, not a floating-point eigenvalue treated as an exact bound.

A concrete continuous-panel enclosure. On a panel contained in $|s - s_c| \leq h$, put $Q_c = \mathcal{F}_0(s_c)^{-1}$, $W_c = Q_c \mathcal{U}$, and

$$e_\tau(h) = \max_i e^{-\Re(s_c)\tau_i} (e^{h\tau_i} - 1).$$

For fixed delays,

$$\|\mathcal{F}_0(s) - \mathcal{F}_0(s_c)\|_2 \leq h + \|B_0\|_2 e_\tau(h), \quad \|\mathcal{V}(s) - \mathcal{V}(s_c)\|_2 \leq \sqrt{2} e_\tau(h).$$

If verified bounds give $a_c := \|Q_c\|_2[h + \|B_0\|_2 e_\tau(h)] < 1$, the Neumann series yields

$$\|T(s) - T(s_c)\|_2 \leq \frac{\|\mathcal{V}(s_c)\|_2 a_c + \sqrt{2} e_\tau(h)}{1 - a_c} \|W_c\|_2 =: \epsilon_{T,c}. \quad (18)$$

Hence $R_d(|T(s_c)| + \epsilon_{T,c} \mathbf{1}\mathbf{1}^T)$ is an entrywise panel envelope. Verified rounding/solve errors in $Q_c, W_c, T(s_c)$ must be added; these formulas do not make an ordinary dense solve self-certifying. One positive-vector test on that envelope bounds the entire panel and the entire parameter box. A finite covering establishes this contour condition. The displayed constants apply in these coordinates; any nonunit state/equation scaling requires rederiving the perturbation bounds in the scaled coordinates. Thus the outstanding computation has an explicit error formula, rather than an unspecified Hessian-sampling step.

3.5 Regional maximum and a controller-eliminated witness

Status: PROVED_REDUCED_MODEL. Stack the necessary cluster inequalities and their verified errors. Let $d = (\delta\rho, \delta K)$ range over compact boxes $\mathcal{R} \times \mathcal{K}$, intersected with the original physical limits. Let $A = J_\rho$, $C = J_K$, $b = -g_0 + \beta$. Every fully secure design in this region necessarily satisfies

$$A\delta\rho + C\delta K \leq b.$$

Consequently the LP

$$U_{\mathcal{D}} = \max_{\delta\rho \in \mathcal{R}, \delta K \in \mathcal{K}} (P^0)^T \delta\rho \quad \text{subject to } A\delta\rho + C\delta K \leq b \quad (19)$$

gives

$$\boxed{P_F \leq P_{\mathcal{D}}^* \leq (P^0)^T \rho_0 + U_{\mathcal{D}} = P_{U,\mathcal{D}}.} \quad (20)$$

The lower candidate must belong to this same region and pass the full spectrum and all declared events. Numerically validated feasibility remains numerical evidence unless its own errors are certified. Omitting frequency, RoCoF or actuator constraints from this LP is an optimistic relaxation, not a safety claim.

For any $\eta \geq 0$, the support functions of the two boxes give

$$U_{\mathcal{D}} \leq \eta^T b + h_{\mathcal{R}}(P^0 - A^T \eta) + h_{\mathcal{K}}(-C^T \eta). \quad (21)$$

Minimization over η recovers the feasible bounded LP value by ordinary LP duality. For an interval box, $h(z) = \sum_j \max(l_j z_j, h_j z_j)$. The bound is valid for *all* replacement directions in the declared region, unlike the earlier fixed-ray calculation.

For a proposed replacement $\delta\rho$, the strict inequality

$$\eta^T (A\delta\rho - b) > h_{\mathcal{K}}(-C^T \eta) \quad (22)$$

proves that no permitted local gain vector in the region can satisfy the necessary conditions, hence none can meet the full security contract. The weights identify which collective requirements conflict. If $C^T \eta = 0$, their first-order aggregate has no gain authority; finite residual authority is still allowed by $\eta^T \beta$. Only the strict finite inequality, not a small derivative alone, establishes exclusion. This is a regional statement, not a global impossibility theorem for all local controllers.

3.6 Where graph interactions actually enter

Status: EXACT_IDENTITY. Group the rows/columns of H by intervention site (two algebraic voltage-equation channels and two gain channels per candidate). The exact series contains

$$\text{tr } H^2 = \sum_i \text{tr } H_{ii}^2 + \sum_{i \neq j} \text{tr}(H_{ij}H_{ji}),$$

and $\text{tr } H^\ell$ sums closed walks of length ℓ through these dynamical intervention ports. Their weights include the full resolvent, P/Q coupling, SG dynamics and exact delay phases. They are not necessarily physical transmission-line cycles, and H is neither a Laplacian nor a physical damping matrix. Even its diagonal blocks depend on the full network.

Writing $H = D + O$ with D block diagonal gives

$$\det(I + H) = \det(I + D) \det(I + T), \quad T = (I + D)^{-1}O, \quad \text{tr } T = 0.$$

When the relevant logarithm series converge, the first omitted inter-site term after discarding O is $-\frac{1}{2} \text{tr } T^2$ in the log determinant. Its contribution to the root-sum change has the opposite sign because of (5); it need not be stabilizing. If $\|T\|_2 \leq t < 1$, the discarded log-determinant magnitude is at most $r\varphi(t)$. This supplies a falsifiable approximation error for neglecting collective interactions, rather than assuming graph roughness predicts damping.

An orthogonal graph basis can reorganize these operators. It preserves traces and Euclidean norms for the corresponding similarity, but its diagonal truncation changes the approximation. Low graph frequency alone does not justify that truncation. No GSP advantage or physical cycle mechanism is established until the retained/off-diagonal terms explain a measured limiting decision.

4 Collective curvature and a graph-resolved finite remainder

Status: EXACT_IDENTITY. Let $d \in \mathbb{R}^{3n}$ denote ordinary replacement and gain increments. Define fixed diagonal selectors E_a such that $\Theta(d) = \sum_a d_a E_a$: a replacement selector contains two unit entries, whereas a gain selector contains one. In the affine descriptor chart, $H(s, d) = \sum_a d_a E_a T(s)$ exactly. All delays remain fixed inside $T(s)$.

Theorem 1 (Collective cluster curvature). *Under the contour hypotheses of Section 3, the real cluster margin has the second-order expansion*

$$g_\Gamma(p_0 + d) = g_{\Gamma,0} + J_\Gamma d + \frac{1}{2} d^T Q_\Gamma d + \mathcal{R}_{\Gamma,3}(d), \quad (Q_\Gamma)_{ab} = \Re \frac{1}{2\pi i m_\Gamma} \oint_\Gamma \text{tr}(E_a T E_b T) ds. \quad (23)$$

The matrix Q_Γ is real symmetric. It is not required to be positive semidefinite and is not a Laplacian.

Proof. Substitute $H = \sum_a d_a E_a T$ in the convergent series for $-\text{tr} \log(I + H) = -\text{tr } H + \text{tr } H^2/2 - \dots$ in (5). Cyclic invariance of the trace gives $\text{tr}(E_a T E_b T) = \text{tr}(E_b T E_a T)$. Collecting terms proves the formula. Uniform convergence on the compact contour permits termwise integration. This differentiates a cluster moment, so internal root collisions do not require eigenvector derivatives or a simple-root assumption. $\square$

For scalar selectors the integrand is $T_{ab}T_{ba}$. Thus a mixed change at two sites follows a return path through the full dynamical network. Two individually favorable first-order changes need not remain favorable when combined. The sign is set by the complex contour integral, not by graph adjacency or a positive edge weight. Diagonal terms already include the full grid.

Status: PROVED_REDUCED_MODEL. Suppose a continuous, piecewise bounded nonnegative envelope satisfies

$$|H(s, d)| \leq M(s), \quad \sup_{s \in \Gamma} \rho(M(s)) < 1, \quad d \in \mathcal{D}_d.$$

A verified positive-vector inequality on every panel is one sufficient way to establish this condition. Set

$$\Psi_k(M) = \sum_{\ell=k}^{\infty} \frac{\text{tr } M^\ell}{\ell} = -\log \det(I - M) - \sum_{\ell=1}^{k-1} \frac{\text{tr } M^\ell}{\ell}, \quad k \geq 2.$$

Here the determinant is positive and the logarithm is real: real eigenvalues of M are below one, and nonreal eigenvalues occur in conjugate pairs. Despite that spectral representation, every term in the defining trace series is nonnegative.

Theorem 2 (Closed-walk remainder and interaction-omission budget). *The remainder after degree $k - 1$ obeys*

$$|\mathcal{R}_{\Gamma,k}(d)| \leq \beta_{\Gamma,k}^{\text{walk}} := \frac{1}{2\pi m_\Gamma} \oint_{\Gamma} \Psi_k(M(s)) |ds|. \quad (24)$$

Partition channels into physical intervention sites and let $H_b = \text{blockdiag}(H_{11}, \dots, H_{nn})$. The omitted contribution to the cluster-shift formula, obtained by replacing H with H_b , is bounded by

$$\beta_{\Gamma}^{\text{cross}} = \frac{1}{2\pi m_\Gamma} \oint_{\Gamma} \left[\Psi_2(M) - \sum_i \Psi_2(M_{ii}) \right] |ds|. \quad (25)$$

This is a bound on an algebraic approximation to the contour formula; it does not assert that H_b is a realizable physical grid.

Proof. The entrywise inequality $|H^\ell| \leq M^\ell$ implies $|\text{tr } H^\ell| \leq \text{tr } M^\ell$. Sum the tail of the logarithm series and integrate its absolute value to obtain (24). For (25), the terms remaining after block-diagonal subtraction are exactly index sequences that visit more than one block before returning to their starting channel. Bounding the absolute value of each product by the corresponding product of entries of M gives $\text{tr } M^\ell - \sum_i \text{tr } M_{ii}^\ell$. The degree-one difference is zero. Summation proves the result. $\square$

The earlier scalar bound follows from $\text{tr } M^\ell \leq r\kappa^\ell$ whenever a common induced norm of M is bounded by $\kappa < 1$. Retaining the actual walk sums can therefore substantially reduce conservatism. A computable, slightly looser expression avoids logarithms:

$$\Psi_k(M) \leq \frac{1}{k} \text{tr} \left[M^k (I - M)^{-1} \right].$$

All inverse, determinant, integration and rounding errors must still be enclosed. In particular, a verified evaluation of (25) requires an *upper* enclosure of the total minus *lower* enclosures of the within-site terms; subtracting two upper bounds would be invalid. Directly summing cross-site walks is an alternative.

Meaning of beyond nodal damping. Equation (23) identifies the signed, frequency-dependent interaction between two replacement or tuning actions. Equation (25) quantifies the error made by dropping those interactions. These are not nodal damping constants: T contains the grid resolvent, SG dynamics, voltage/PQ coupling and the exact phases $e^{-s\tau_i}$. They supplement the physical torque-speed damping operator derived in the appendices; they do not turn a contour return matrix into a physical impedance. The closed walks are paths in the intervention-port representation, not automatically cycles of transmission lines. No communication between PLLs has been introduced.

4.1 A stronger necessary replacement relaxation

Write $d \in [l, u]$ and introduce a real symmetric matrix Z . Every actual design admits $Z = dd^T$, hence satisfies

$$\begin{bmatrix} 1 & d^T \\ d & Z \end{bmatrix} \succeq 0, \quad (26)$$

$$Z_{ab} \geq l_a d_b + l_b d_a - l_a l_b, \quad Z_{ab} \geq u_a d_b + u_b d_a - u_a u_b, \quad (27)$$

$$Z_{ab} \leq u_a d_b + l_b d_a - u_a l_b, \quad Z_{ab} \leq l_a d_b + u_b d_a - l_a u_b. \quad (28)$$

Retain the diagonal case of these inequalities as well. Impose, for each verified cluster,

$$g_{\Gamma,0} + J_{\Gamma}d + \frac{1}{2}\langle Q_{\Gamma}, Z \rangle \leq \beta_{\Gamma,3}^{\text{walk}}. \quad (29)$$

Maximizing $(P^0)^T \delta \rho$ over this semidefinite relaxation gives an optimistic regional replacement upper bound. Dropping $Z = dd^T$ enlarges the feasible set, which is the required direction; the returned gains need not be physically feasible. Neither a primal floating-point optimum nor a tiny solver residual is a certified upper bound. A feasible dual with enclosed errors, or another verified objective bound, is required for that label.

Certified coefficient errors add $e_0 + \sum_a e_{J,a} r_a + \frac{1}{2} \sum_{a,b} e_{Q,ab} r_a r_b$ to the right-hand side, where $r_a = \max(|l_a|, |u_a|)$. The lift may sharpen the linear relaxation, but no universal dominance follows if different remainders or additional relaxations are used. Both outer problems can be intersected because each contains all secure designs. This strengthening has been derived and identity-tested; the IEEE-39 semidefinite upper-bound problem has not been solved.

5 A reproducible analytical design and bound procedure

The optimization and certification roles are distinct. The exact all-PLL formula in Eq. (59) constructs candidates. The regional outer problem permits arbitrary admissible modal patterns and limits what any such candidate can achieve.

1. Freeze the sharing convention, fixed delay vector, physical gain limits, replacement region, event set, and output definitions. Solve and audit the equilibrium; verify the algebraic chart.
2. At a candidate center, assemble the exact descriptor with exponential delay factors. Use numerical roots to propose contours; verify their counts, boundary regularity and parameter-box envelopes. Numerical root locations alone do not close this step.
3. Integrate the reference moments and J_{Γ} (and, if used, Q_{Γ}) with error enclosures. Evaluate the walk remainder. If a panel or algebraic chart cannot be verified, subdivide it or return an unresolved box; do not fill it by interpolation.
4. Solve the necessary LP or lifted outer problem and verify an upper objective bound or a strict controller-exclusion witness. Compare this upper bound against the box-only replacement ceiling.
5. Use the analytical gain map or a trust-region local step to propose a design, then evaluate the full delayed spectrum and all events. Failed designs do not supply a feasible lower bound.
6. Report the same-domain feasible and upper values separately, with their uncertainty. Subdivide parameter regions only under a declared branch rule if a wider bound is sought.

No global convergence theorem for the predictor/corrector is claimed. A global upper bound would require a verified cover of the full design domain, including all support charts and unresolved boxes. The full event-feasible problem generally has more active constraints than the two-row reduced LP; its SG-retention structure cannot be inferred from a two-anchor theorem.

6 Targeted validation of the all-PLL construction

Status: `NUMERICALLY_VALIDATED`. This subsequent experiment tests Eq. (59) with one preregistered replacement step. It is not an optimization campaign. All ten fractions change from $\rho_i = 0.875$ to 0.876 with $\tau_i = 40$ ms fixed. The target is the leading PLL-family pole in the frozen 2–12 Hz catalog, not the globally rightmost pole:

$$\lambda_\star = -0.80156077 + i30.85047169 \text{ s}^{-1}.$$

Its PLL-angle pattern is held fixed and all twenty gains are calculated from the full-grid return. The trial design and modal predictions were hashed before independent full-model validation.

Known gains were reconstructed for all eleven nonreal catalog roots, with maximum relative error 7.266×10^{-8} . Nine vector derivative tests, using three modes and buses 30, 34 and 38, had maximum relative error 2.629×10^{-5} and maximum absolute infinity error 2.965×10^{-1} . The largest hidden-block condition number was 1.637×10^9 ; these residuals establish numerical consistency, not physical accuracy to all the reported digits. Direct automatic differentiation of the nonlinear equations gave maximum relative Jacobian discrepancy 3.682×10^{-18} and maximum equilibrium infinity residual 2.807×10^{-9} . This cross-check uses the same nonlinear device equations; it is not an independent-simulator or physical-data validation.

An independently assembled Jacobian of the same model, followed by the delayed calculation, preserved the target pole within $1.511 \times 10^{-13} \text{ s}^{-1}$ with PLL-pattern MAC 1.000000000000. For the other modes, the largest relative finite-step prediction error was 2.791%; the preregistered 1% criterion concerns derivative validation, not a finite-step Taylor remainder bound. A numerical argument-principle count, cross-checked against determinant phase and closed by a tail bound after rotational deflation, found no roots with $\Re s > -0.05 \text{ s}^{-1}$ for the baseline, retuned design, or fixed-gain comparator. Exact exponentials were used; floating-point contour counts are not interval certificates.

The retuned design passed all five frozen nonlinear events, simulated with the canonical method of steps (Rodas5P, tolerance 10^{-9} , maximum step 0.01 s, horizon 60 s). The 39-bus phase metrics use 0.5 s windows. Frequency and RoCoF limits are 0.5 Hz and 0.5 Hz/s; normalized SG actuator slack must exceed 0.002.

Bus	Load change [MW]	$ \Delta f $ [Hz]	RoCoF [Hz/s]	SG slack
8	-100	0.465412	0.138451	0.096623
16	+100	0.447157	0.126491	0.006341
16	-100	0.474624	0.129044	0.093434
29	+100	0.431226	0.128104	0.011190
29	-100	0.444619	0.200774	0.102894

All voltage samples remained within $[0.972062, 1.078775]$ pu against the declared $[0.9, 1.1]$ limits. The candidate contains 4732.818715 MW of GFL (87.600%) and 669.942375 MW of retained SG: a prescribed increase of 5.402761 MW. The full vectors are in `experiments/all_pll_gain_map_validation_20261003/designs/analytic.toml`.

Interpretation and negative evidence. This closes the numerical chain from an analytical gain construction to a feasible full-model candidate for the declared events. It does not establish a maximum or demonstrate that retuning is needed for this step: the same replacement with unchanged gains also passes the spectral margin. The subsequent matched nonlinear comparison in Section 8 finds that both designs pass all five events, with small opposing changes in frequency and RoCoF metrics. Neither superiority nor exact performance equivalence is established. Two frozen eigenpairs/patterns produce nonzero assignment residuals after replacement, but this does not prove that local PLLs cannot stabilize the network when their modal patterns are allowed to change. No heterogeneous-delay, neighboring-feedback, GSP-compression, convergence, or global-optimality claim follows from this trial. Data, prediction hashes, code, trajectories and claim limits are in `experiments/all_pll_gain_map_validation_20261003/`.

Reproducible gain vector. For buses 30–39, every listed replacement fraction is $\rho_i = 0.876$. These are constructed feasible-candidate gains, not optimal gains.

Bus	$K_{p,i}$	$K_{i,i}$
30	28.27608095	246.76647162
31	28.24652453	245.80547706
32	28.24503605	245.70201441
33	28.21712688	244.76013817
34	28.17474906	246.56932577
35	28.24738622	245.39990024
36	28.25363221	246.19628049
37	28.27912035	246.57391812
38	28.25524609	245.98556358
39	28.28100338	246.76474736

7 Bound checks, falsification and regional evidence

Status: NUMERICALLY_VALIDATED. The descriptor determinant and affine-factor identities were checked at the three frozen frequencies 0.5, 5 and 10 Hz, with $s = -0.05 + i2\pi f$, heterogeneous delays uniformly spread from 0 to 40 ms, replacement displacement radii 0.001, and ordinary gain radii 1%. This delay pattern is different from the uniform-delay nonlinear trial; their lower and upper evidence cannot be mixed. The largest previously recorded normalized identity discrepancy was 1.011×10^{-12} .

The new checks retain those same points and box. They compare the scalar logarithm-tail bound with the entrywise closed-walk majorant. The table contains pointwise bounds on the trace logarithm, not eigenvalue shifts or a sampled replacement certificate.

f [Hz]	scalar tail	walk tail	reduction factor
0.5	0.0189751	0.000964533	19.67
5	1.38717	0.0303027	45.78
10	0.0056687	0.000313074	18.11

The first- and second-order trace predictions and the cross-site omission budgets passed all three pointwise numerical checks. This supports improved bound sharpness at these points; it does not demonstrate that graph terms increase attainable replacement.

For the separate two-state pure-DDE fixture

$$\Delta_{\text{toy}}(s, p) = \text{diag}(s + 1, s + 2) - pe^{-s/10} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad |p| \leq 0.01,$$

the circle centered at -1 with radius $1/4$ admits the analytic whole-circle bound $\kappa \leq 12/175 < 1$ and one enclosed root. Implicit differentiation gives $\lambda'(0) = e^{1/10}$ and $\lambda''(0) = 1.8e^{1/5}$. The contour Hessian reproduces both values to numerical error. For the five frozen values of p at 256 and 512 quadrature nodes, the largest quadratic error is 3.140×10^{-6} , below the analytic cubic-tail bound 5.770×10^{-5} . This is a theorem/check on the declared fixture, not on the power grid.

7.1 Continuous IEEE-39 contour test

Complex ball arithmetic [23] at 128-bit precision encloses every point of a square centered at the preserved slow-mode estimate $-0.12229688588491999 + i0.32408378548166134$, with half-side 0.04. The region uses uniform fixed delay 40 ms, $\rho_i \in [0.874, 0.876]$ and ordinary gains within 1% of the baseline. The matrix inputs are the exact binary64 values parsed from the frozen exports. The fixed delay is the exact decimal $0.04 = 1/25$ s enclosed in ball arithmetic, not its binary64 rounding. Physical and equilibrium uncertainty are not enclosed by this numerical certificate. An interval inverse verifies $G(\rho)$ throughout that box. The contour cover uses 485 accepted panels and gives $\kappa \leq 0.98794235 < 1$. The enclosed argument-principle integral contains exactly one integer, 1. Consequently one root remains inside this square for every permitted parameter combination. Its real part is below -0.08229 s^{-1} , so this one cluster satisfies the -0.05 s^{-1} margin throughout the box. This verifies one cluster; it is not a complete-spectrum or nonlinear certificate. All ball enclosures, parameters and code are saved in the reproduction package. The coarse first-order scalar moment remainder is bounded by 4.458179 s^{-1} , much larger than the required margin. This bound is conservative even though the contour geometry proves this cluster safe. The sharper walk majorant has not yet been integrated with verified errors around this contour.

7.2 A decisive negative result about the chosen regional box

Status: `NEGATIVE.RESULT`. To the reported numerical precision, the feasible 87.6% design has $\rho_i = 0.876$ at every candidate and its gains lie inside the uniform-delay verification box. Its reported objective coincides with the box-only upper bound

$$P_{\text{box}} = 0.876(P^0)^T \mathbf{1} = 4732.818715 \text{ MW}.$$

Conditional on true feasibility of the exact upper-corner design, this is the optimum *of that box*, by monotonicity of a positive dispatch objective. No spectral theorem is needed for this fact. It is not an irreducible SG-support bound and is not a meaningful near-global-optimality result. A purported upper bound below this value would contradict the numerical witness and require resolving that discrepancy. For strict arithmetic bookkeeping, the binary64 literal 0.876 exceeds the exact decimal endpoint by $1/1125899906842624000$; the numerical witness is not treated as an interval-certified corner point. No rigorous lower-upper gap is claimed from these rounded values. Thus even perfect verification of this regional box cannot establish the main application claim of a physical interior replacement limit. The manuscript does not expand the box after observing this result.

8 Targeted closure: physical damping and matched nonlinear comparison

This follow-up adds two declared tests without changing the historical design, events, gains, or frozen result files. Its protocol and outputs are stored in `experiments/poster_julia_validation_20261003/`. The harmonic tests are algebraic; the event comparator uses Julia 1.11.9.

8.1 Finite-delay physical damping law in IEEE-39

Status: NUMERICALLY_VALIDATED. For the historical baseline and analytical designs, at each of $f = 0.5, 5, 10$ Hz, increase the delay at one of buses 30–39 by exactly the numerical increment 1 ms, starting from uniform 40 ms. All $2 \times 3 \times 10 = 60$ cases are retained. No case was chosen by its sign or stability outcome. At each point, independently eliminate the 194 hidden states in the old and modified 204-state operators and compare the difference with the finite rank-one formula in Appendix B. The physical mass scaling is $M_i = 2H_i S_i^0(1 - \rho_i)$, in MWs for per-unit speed.

The largest relative error in $\delta Z = \gamma_i u_i v_i^*$ is 4.85×10^{-8} . Every computed Hermitian increment has one positive and one negative eigenvalue above the declared relative threshold $10^{-7} \|\delta D_H\|_2$; the other eight eigenvalues have relative magnitude at most 1.40×10^{-9} . The minimum squared sine between the two propagation paths is 0.00327, so none of these sampled paths is numerically collinear. Hidden-block condition numbers and all signed eigenvalues are reported in `TABLE_02.FINITE_DELAY_RANK_LAW.csv`. This supports the finite physical identity at the declared points. It does not certify an interval of frequencies, imply stability from the sign of the increment, or improve a replacement-capacity bound.

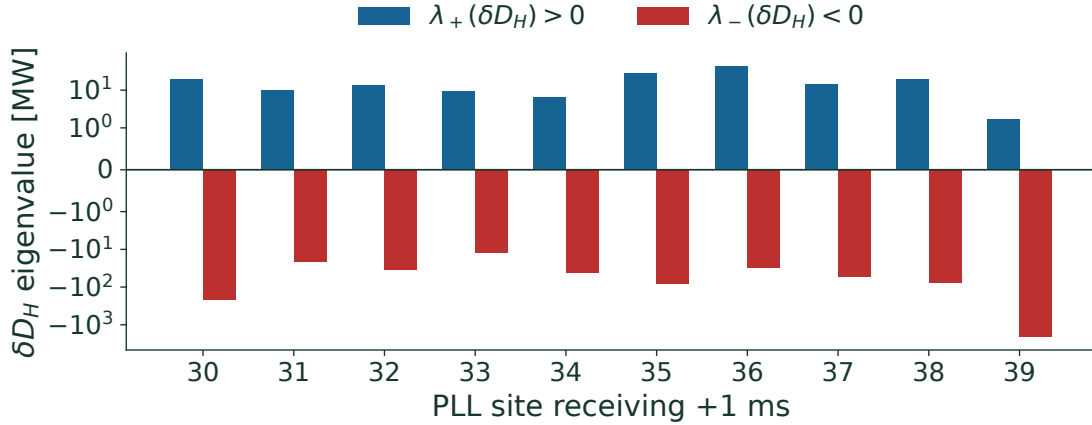


Figure 1: Physical damping redistribution at the preregistered illustrative frequency 5 Hz, analytical design. One PLL delay is increased from 40 to 41 ms at a time. The symmetric-log scale exposes both signs. This is a harmonic linear-model check, not a nonlinear stability test.

8.2 Same replacement, unchanged versus analytical gains

Status: NUMERICALLY_VALIDATED. The missing matched comparator is now executed at the same $\rho_i = 0.876$ and $\tau_i = 40$ ms as the analytical candidate. Its gains remain at the common baseline values. The canonical method-of-steps solver uses Rodas5P, tolerance 10^{-9} , maximum step 0.01 s, and a 60 s event horizon. Both phase-derived frequency and RoCoF metrics use the canonical 0.5 s window over all 39 bus voltages. The five nominal event magnitudes correspond to the existing constant-impedance perturbation contract. Equilibrium residual and gain bounds are checked before execution.

Both designs pass all five events. Analytical retuning increases frequency excursion by 0.000177–0.000207 Hz and reduces RoCoF by 0.0000744–0.000435 Hz/s across these cases. Its SG actuator slack is also slightly smaller. These are small numerical tradeoffs, not a demonstrated engineering advantage; no refinement study was performed to turn these small differences into accuracy-certified performance claims. The purpose of the exact gain map is achieved: the assigned PLL eigenpair is

Table 1: All five matched events. F is peak absolute frequency deviation in Hz; R is peak absolute RoCoF in Hz/s. A denotes the analytical map; fixed retains the original gains. All original guards also pass.

Bus, MW	F_{fixed}	F_A	R_{fixed}	R_A	Guards
8, -100	0.465209	0.465412	0.138636	0.138451	Pass / Pass
16, +100	0.446979	0.447157	0.126583	0.126491	Pass / Pass
16, -100	0.474417	0.474624	0.129118	0.129044	Pass / Pass
29, +100	0.431049	0.431226	0.128359	0.128104	Pass / Pass
29, -100	0.444418	0.444619	0.201209	0.200774	Pass / Pass

preserved after the prescribed replacement step. This matched trial does not show that retuning was necessary for feasibility, that it maximized replacement, or that it dominates fixed gains.

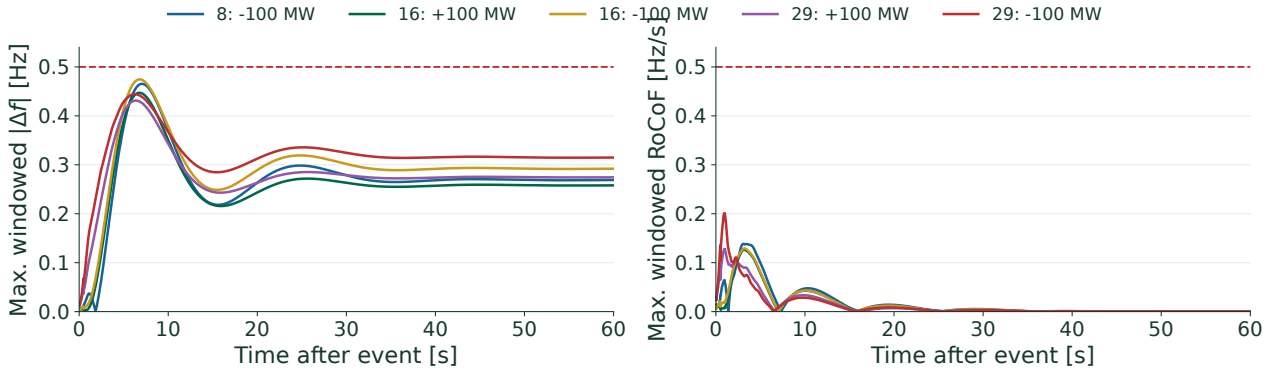


Figure 2: Preserved analytical-design trajectories for every frozen event, plotted across the full 60 s horizon. Frequency and RoCoF use the declared 0.5 s phase window. The new comparator’s complete trajectories and paired metrics are included separately in the archive.

Updated contribution boundary. The physical finite-delay theorem, the collective contour Hessian, and the walk remainder provide explicit equations for network-mediated effects. The numerical evidence now includes all 60 physical-law checks and a matched five-event comparator. No informative verified physical replacement maximum follows. The posterior comparison therefore supports a theory-and-validation poster; the stronger application claim remains open.

9 Discussion, contribution boundary and limitations

Result	Evidentiary scope
Affine descriptor and gain map	Exact identities in the fixed chart.
Cluster Hessian and walk tails	Proved under the stated contour/envelope hypotheses.
Regional LP / lifted outer bound	Necessary relaxation; verified data and dual bound required.
87.6% candidate, five events	Full-model numerical result for uniform 40 ms delay.
18–46-fold tail reduction	Three frozen algebraic points, not a capacity increase.
Physical maximum or global gap	Not established.
Local PLLs fundamentally insufficient	Not established with freely varying modal patterns.
GSP compression or physical cycle law	Not established.

What is technically added. The model-specific lift links replacement and PLL gain increments to a finite characteristic perturbation without a network-inverse parameter remainder. Cluster curvature resolves pairwise collective actions, and the walk budget quantifies the error of dropping them. The outer relaxation allows all admissible gains and modal patterns; a strict finite dual witness would therefore be stronger than failure to preserve a prescribed eigenvector. This is an applied bound construction assembled from established tools. Bibliographic priority for the combination has not been proved.

What still prevents a strong application paper. An informative verified regional upper bound or controller-exclusion witness below the box-only ceiling is missing. The matched nonlinear fixed-gain comparison now shows both designs feasible, with small opposing metric changes. A fair fixed-structure spectral-abscissa/SQP comparator with identical constraints remains missing. A controller tuned with $I + L + L^2$ terms has not been implemented or shown necessary. The graph Fourier basis may express interactions, but no low-frequency truncation theorem follows.

Physical limits. The certificate concerns a local linearization of a specific nonlinear model. Nonlinear event tests remain separate; no global nonlinear stability, arbitrary-disturbance safety, or unmodeled converter-current-limit guarantee follows. Delays may be heterogeneous in the derivations, but the preserved nonlinear feasible result has uniform delays. The graph moment in the appendices concerns a declared swing-network class and has not been transferred to the full IEEE–39 converter model.

10 Conclusion

Collective effects can be made explicit at the level of finite parameter changes: the full delayed network produces a cluster Hessian and a closed-walk remainder, which lead to necessary regional replacement bounds. The derivations permit moving modes and fixed heterogeneous delays without a low-frequency delay approximation. The available IEEE–39 results validate algebraic sharpness and one feasible analytical retuning construction. One continuous complex-ball calculation also certifies retention and margin stability of a slow cluster over its complete local box. These results do not yet establish a physically limiting replacement ceiling or superior co-design performance. The next decisive result is an informative bound on a preregistered region extending beyond the current feasible corner, followed by a comparison under the same nonlinear security contract.

Reproducibility and research integrity

The main source is edited in place; pre-edit source bytes and prior experiment artifacts are preserved. New checks, protocol, exact input copies, package versions, tables, claim ledger and manifests are under `experiments/regional_paper_closure_20261003/`. Numerical table entries in the new main text are generated from the saved result files. The initial closure performed linear-algebra and contour checks. The subsequent poster closure adds the 60 physical-law checks and five new fixed-gain nonlinear comparator trajectories under `experiments/poster_julia_validation_20261003/`. The historical event evidence remains explicitly attributed to its frozen experiment. Three independent reviews audited the equations, evidence provenance and literature boundary; their corrections and unresolved objections are recorded in the package.

A First correction: derive the closed PLL before assigning damping

Status: EXACT_IDENTITY. At a locked equilibrium $\theta_i^* = \arg V_i^*$, let $V_i^\circ = |V_i^*| > 0$ and let ϕ_i be terminal phase perturbation. The linearized detector is $\delta e_i = V_i^\circ(\phi_i - \theta_i)$. Eliminating the PLL frequency and integrator gives, for zero history,

$$H_i(s) = \frac{\theta_i(s)}{\phi_i(s)} = \frac{V_i^\circ(K_{p,i}s + K_{i,i})e^{-s\tau_i}}{s^2(1 + t_{f,i}s) + V_i^\circ(K_{p,i}s + K_{i,i})e^{-s\tau_i}}. \quad (30)$$

For $K_{i,i} > 0$, direct Taylor division proves

$$H_i(s) = 1 - \frac{s^2}{V_i^\circ K_{i,i}} + \frac{K_{p,i}/K_{i,i} - t_{f,i} - \tau_i}{V_i^\circ K_{i,i}} s^3 + O(s^4). \quad (31)$$

Pure delay first appears at cubic order in this phase-tracking transfer, not in its constant, linear or quadratic coefficients. Any subsequent reduction using analytic factors and inverses nonsingular at zero preserves the $O(s^3)$ delay difference. Dividing an angle operator by s shifts that difference to $O(s^2)$. Singular elimination can change the order. The previously ill-conditioned full hidden block at $s = 0$ prevents an unqualified application of this argument to the full reduced grid.

Consequently $L_\tau = \text{Sym}(B_c K_i T_\tau C)$ is not established for this implemented closed PLL. It may arise for another loop or approximation, which would need its own derivation. At finite frequency the inverse tracking defect is exactly

$$Q_i(s) = H_i(s)^{-1} - 1 = \frac{s^2(1 + t_{f,i}s)e^{s\tau_i}}{V_i^\circ(K_{p,i}s + K_{i,i})}, \quad (32)$$

$$|Q_i(i\Omega)| = \frac{\Omega^2 \sqrt{1 + \Omega^2 t_{f,i}^2}}{V_i^\circ \sqrt{K_{i,i}^2 + \Omega^2 K_{p,i}^2}}, \quad (33)$$

$$\arg Q_i(i\Omega) = \pi + \arctan(\Omega t_{f,i}) - \arctan(\Omega K_{p,i}/K_{i,i}) + \Omega\tau_i \pmod{2\pi}. \quad (34)$$

Delay rotates this defect without changing its magnitude. This defect is not itself physical damping or a Laplacian.

B Physical collective damping and its finite latency law

Status: EXACT_IDENTITY. The exact characteristic operator of the full linearization is

$$\Delta(s) = sE - A_0 - \sum_i b_i c_i^T e^{-s\tau_i}.$$

The two delayed PLL injections share one column b_i , with entries $K_{p,i}/t_{f,i}$ and $K_{i,i}$. c_i^T differentiates the actual detector after algebraic voltage elimination.

In exported $E = I$ coordinates, retain per-unit SG speed deviations ν and eliminate all other states z wherever Δ_{zz} is invertible:

$$Z(s) = M(\Delta_{\nu\nu} - \Delta_{\nu z} \Delta_{zz}^{-1} \Delta_{z\nu}), \quad d = Z(s)\nu, \quad D_H(\Omega) = \text{Herm } Z(i\Omega). \quad (35)$$

d_i is mechanical torque expressed as MW at nominal speed, injected as d_i/M_i in the swing equation. For physical harmonic phasors,

$$\langle d^T \nu \rangle = \frac{1}{2} \hat{\nu}^* D_H(\Omega) \hat{\nu}.$$

This is incremental mechanical supply, not total AC power or a nonlinear Lyapunov function. A steady harmonic response needs to exist; an unstable system may only admit a formal particular phasor solution.

Theorem 3 (One delayed channel redistributes physical damping). *Fix ρ , equilibrium, gains, ports and $\Omega > 0$. Change only τ_i to $\tau_i + \delta\tau_i \geq 0$. Assume both hidden blocks are invertible. At $s = i\Omega$, evaluate the following factors at the old design:*

$$\begin{aligned}\kappa_i &= e^{-s\tau_i} - e^{-s(\tau_i + \delta\tau_i)}, & h_i &= c_{z,i}^T \Delta_{zz}^{-1} b_{z,i}, \\ u_i &= M(b_{\nu,i} - \Delta_{\nu z} \Delta_{zz}^{-1} b_{z,i}), & v_i^* &= c_{\nu,i}^T - c_{z,i}^T \Delta_{zz}^{-1} \Delta_{z\nu}, \\ \gamma_i &= \kappa_i / (1 + \kappa_i h_i), & a_i &= \gamma_i u_i.\end{aligned}$$

Then

$$\boxed{\delta Z = \gamma_i u_i v_i^*, \quad \delta D_H = \frac{1}{2}(a_i v_i^* + v_i a_i^*)}. \quad (36)$$

If $\gamma_i \neq 0$ and u_i, v_i are independent, δD_H has exactly one positive and one negative eigenvalue, all others zero.

Proof. The full change is $\kappa_i b_i c_i^T$. Applying Sherman–Morrison inside the Schur complement proves (36). Nonsingularity of both hidden blocks makes $1 + \kappa_i h_i \neq 0$ by the determinant lemma. The range of δD_H is contained in the span of a_i, v_i . Its two possibly nonzero eigenvalues are

$$\ell_{\pm} = \frac{\Re(v_i^* a_i) \pm \sqrt{\|a_i\|^2 \|v_i\|^2 - [\Im(v_i^* a_i)]^2}}{2}.$$

Their product is

$$\ell_+ \ell_- = -\frac{|\gamma_i|^2}{4} (\|u_i\|^2 \|v_i\|^2 - |v_i^* u_i|^2) < 0$$

by strict Cauchy–Schwarz. This proves both signs and rank two. $\square$

This is a finite-change statement for the full linearization, not merely a first-order approximation. It says that a local latency change increases harmonic dissipation in some collective speed directions and decreases it in others, under the stated independence. Hence its negative is not a positive-semidefinite delay penalty either. Collinear paths can instead give a rank-one change of either sign or zero; phase-periodic increments can make $\kappa_i = 0$. Multiple channel changes may combine to have another sign pattern. This theorem alone does not determine stability or nonlinear safety. Section 8 checks path independence numerically at 60 frozen IEEE–39 harmonic cases; no frequency interval is certified.

Differentiating the finite formula gives

$$Z_{\tau_i}(s) = s e^{-s\tau_i} u_i v_i^*. \quad (37)$$

Away from the imaginary axis v_i^* denotes the displayed analytic row, not an additional conjugation of an analytic function.

C Graph meaning of the physical directions

Status: EXACT_IDENTITY. Use a verified symmetric physical Laplacian only to define a basis: $M^{-1/2} L_P M^{-1/2} = U \Lambda U^T$. The lossy model in Z stays intact. Under physical mass normalization,

$$\widehat{\delta D} = U^T M^{-1/2} \delta D_H M^{-1/2} U = \frac{1}{2}(\widehat{a} \widehat{v}^* + \widehat{v} \widehat{a}^*), \quad \widehat{a} = U^T M^{-1/2} a, \quad \widehat{v} = U^T M^{-1/2} v.$$

Thus $\widehat{\delta D}_{k\ell} = \frac{1}{2}(\widehat{a}_k \widehat{v}_\ell + \widehat{v}_k \widehat{a}_\ell)$. Sylvester's law preserves the positive/negative counts under this invertible congruence. Both network paths determine the modal effect; the spatial Fourier spectrum of τ alone cannot supply its sign. Low graph modes are not automatically the right control subspace. The rank-two claim concerns one channel's increment, not the full damping matrix.

D A graph law for inertia and aggregate frequency

D.1 An explicit reduced class

Assume, separately from the full-model theorem,

$$\nu = \dot{q}, \quad Z_r(s)\nu = d, \quad Z_r(s) = sM + \mathcal{D}(s) + L/s, \quad \mathcal{D}(s) = D_0 + sD_1 + O(s^2). \quad (38)$$

M is positive diagonal, $L = BWB^T$ is a connected undirected Laplacian with $W > 0$, and $\mathcal{D}$ is real analytic near zero. For physical per-unit speed, $q = \delta/\omega_0$; consequently $L = \omega_0 L_{\text{angle}}$. M, D_0, L then have units MWs, MW, and MW/s. This scaling is essential. The class (38) has not been derived for the full lossy IEEE-39 PLL/DC model; singular hidden states at zero cannot be ignored.

Define

$$m = \mathbf{1}^T M \mathbf{1}, \quad d_0 = \mathbf{1}^T D_0 \mathbf{1} > 0, \quad u_0 = M^{1/2} \mathbf{1} / \sqrt{m}, \quad M^{-1/2} L M^{-1/2} = U \text{diag}(0, \Lambda_\perp) U^T.$$

Write $\tilde{D} = U^T M^{-1/2} \mathcal{D} M^{-1/2} U$ and $Z_\perp = sI + \tilde{D}_{\perp\perp} + \Lambda_\perp/s$. For inertia-proportional input $d = (M\mathbf{1}/m)p$, the COI speed $w = \mathbf{1}^T M \nu / m$ satisfies

$$z_{\text{coi}}(s) = m[s + \tilde{D}_{00} - \tilde{D}_{0\perp} Z_\perp^{-1} \tilde{D}_{\perp 0}], \quad w = p/z_{\text{coi}}. \quad (39)$$

This is an exact Schur identity for this class wherever defined.

Theorem 4 (Graph correction to the aggregate response moment). *Let $b_R = D_0 \mathbf{1} - (d_0/m)M\mathbf{1}$ and $b_L = D_0^T \mathbf{1} - (d_0/m)M\mathbf{1}$. Then*

$$z_{\text{coi}}(s) = d_0 + s m_{\text{app}} + O(s^2), \quad \boxed{m_{\text{app}} = m + \mathbf{1}^T D_1 \mathbf{1} - b_L^T L^\dagger b_R}. \quad (40)$$

If D_0 is symmetric, $b_L = b_R = b$ and

$$\boxed{b^T L^\dagger b = m \sum_{k>0} \frac{\tilde{D}_{k0}(0)^2}{\lambda_k} = \min_{Bf=b} f^T W^{-1} f \geq 0.} \quad (41)$$

Status: PROVED_REDUCED_MODEL.

Proof. $Z_\perp^{-1} = s(\Lambda_\perp + s\tilde{D}_{\perp\perp} + s^2 I)^{-1} = s\Lambda_\perp^{-1} + O(s^2)$. Substitute in (39). Both b_L, b_R sum to zero. The mass-normalized spectral solution of $Lx = b_R$ differs from $L^\dagger b_R$ only by a constant vector, which b_L^T annihilates. This identifies the cross coefficient as $b_L^T L^\dagger b_R$. Symmetry gives the sum of squares. For the flow identity, $f_* = WB^T L^\dagger b$ is feasible and every feasible $f = f_* + c$ has $Bc = 0$. Its cost is $f_*^T W^{-1} f_* + c^T W^{-1} c$, proving the minimum. $\square$

m_{app} is a response moment, not stored kinetic inertia. It may be negative with positive physical inertia in a stable system. The mismatch uses row/column sums of collective damping, not its diagonal alone. Weak graph modes amplify the effect by $1/\lambda_k$. Cycle flows belong to $\ker B$ and add the nonnegative cost above; no separate cycle damping object is needed. At fixed b , $\partial(b^T L^\dagger b)/\partial w_e = -(B_e^T L^\dagger b)^2 \leq 0$. This concerns the moment at a fixed operating point, not monotonic improvement of nadir or stability under actual line changes.

Corollary 5 (Negative moment forces an overshoot). *Suppose the frequency response is stable with integrable transient. For $p(t) = -P1_{t \geq 0}$, $P > 0$, define signed frequency drop $x = -w$. Then $x_\infty = P/d_0$ and*

$$\boxed{\int_0^\infty (x_\infty - x(t)) dt = P m_{\text{app}} / d_0^2.} \quad (42)$$

If $m_{\text{app}} < 0$, the nadir is below the final frequency.

Proof. Expand $X(s) = P/[sz_{\text{coi}}(s)]$. The limit of $P/(d_0 s) - X(s)$ at zero is the stated area. A negative integral cannot have an everywhere nonnegative integrand. $\square$

A positive moment does not rule out overshoot. A negative one does not quantify the peak or establish violation of a declared limit.

Exact two-node example; no time simulation. Let $M = I_2$, $L = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ in dimensionless units. Both $D_0 = 5I_2$ and $D_0 = \text{diag}(1, 9)$ have $m = 2, d_0 = 10$. The former has $w/p = 1/(2s + 10)$, monotone. For the latter, $b = (-4, 4)^T$, $b^T L^\dagger b = 16$, $m_{\text{app}} = -14$, and

$$\frac{w}{p} = \frac{s^2 + 5s + 2}{2(s^3 + 10s^2 + 11s + 10)}.$$

The cubic is Hurwitz by $10, 11, 10 > 0$ and $10 \cdot 11 > 10$. Equation (42) proves overshoot. Equal total inertia and total damping do not determine the frequency transient. This example is not an IEEE-39 result.

For a general deficit input $d(t) = -v1_{t \geq 0}$ with $\mathbf{1}^T v = P > 0$, put $a = v - (P/m)M\mathbf{1}$. Eliminating the disagreement input yields

$$\int_0^\infty (x_\infty - x(t)) dt = \frac{P m_{\text{app}}}{d_0^2} + \frac{b_L^T L^\dagger a}{d_0}. \quad (43)$$

The extra term comes from the numerator $P - sb_L^T L^\dagger a + O(s^2)$. Thus the equations accept arbitrary input location. Algebraic load events need their real input map, not an assumed mechanical step.

E Nonlinear causality, RoCoF and replacement

Proposition 6 (A causal limitation on PLL retuning). *Assume the nonlinear DDAE is well posed, its pre-event detector history is zero, its trim is gain-independent, and gains occur only as in (3). At fixed other parameters, changing site i 's PLL gains cannot change the physical trajectory for $0 \leq t < \tau_i$. If all tunable channels have positive delay, the trajectory before $\tau_{\min}$ is independent of all their gains.*

Status: EXACT_IDENTITY.

Proof. The corresponding delayed error is zero on the stated interval. Both designs have the same vector field and history there; uniqueness of the retarded initial-value problem gives equality. $\square$

Other converter states, currents, DC dynamics, SG governors and algebraic voltages can respond immediately. This is not a claim that GFL power is frozen before the PLL delay. Any output-guard failure in that common interval is unrepairable by these gains. If $\mathcal{S}_{\text{blind}}$ is the set of ρ passing all declared event/output guards on that interval, then

$$\{\rho : \exists K \text{ fully feasible}\} \subseteq \mathcal{S}_{\text{blind}}, \quad P_{GFL}^* \leq \sup_{\rho \in \mathcal{S}_{\text{blind}}} \sum_i P_i^0 \rho_i.$$

This is an unevaluated necessary-condition relaxation, potentially weak for short delays and long measurement windows.

For a mechanical step in (38), with causal controller realization and no instantaneous synthetic-inertia feedthrough,

$$\dot{v}(0^+) = M^{-1}d(0^+), \quad \dot{w}(0^+) = \mathbf{1}^T d(0^+)/m.$$

An instantaneous COI rotor RoCoF limit $R_{\max}$ Hz/s therefore needs

$$\sum_i 2H_i S_i^0 \epsilon_i \geq f_0 |\mathbf{1}^T d(0^+)| / R_{\max}. \quad (44)$$

This classical condition is not novel. Minimizing $P^T \epsilon$ under it and $0 \leq \epsilon \leq 1$ is fractional knapsack: sort $2H_i S_i^0 / P_i^0$ decreasingly for positive dispatch and retain in that order, with at most one final fraction. This gives only the necessary-condition bound for that explicit input/output contract.

The frozen repository instead monitors bus-voltage phase over $T_w = 0.5$ s:

$$F_j(t) = \frac{\phi_j(t) - \phi_j(t - T_w)}{2\pi T_w}, \quad R_j(t) = \frac{\phi_j(t) - 2\phi_j(t - T_w) + \phi_j(t - 2T_w)}{2\pi T_w^2}.$$

Z-load events can jump algebraic voltage phase. These metrics cannot be replaced by instantaneous rotor RoCoF. No numerical upper replacement bound for the frozen problem is computed in this document.

F The exact connection to stability and local design

For a simple root, with nonzero eigenvalue denominator,

$$\lambda_p = -\frac{w^* \Delta_p v}{w^* \Delta_s v}, \quad \Delta_s = E + \sum_i \tau_i b_i c_i^T e^{-s\tau_i}.$$

Delayed memory is not stored inertia. At a regular physical-port root,

$$\lambda_{\tau_i} = -\frac{\lambda e^{-\lambda\tau_i} (\ell^* u_i)(v_i^* r)}{\ell^* Z_s r}. \quad (45)$$

This follows by using (37) in $\lambda_p = -\ell^* Z_p r / (\ell^* Z_s r)$. Phase and left/right mode overlaps matter; the harmonic sign theorem does not replace root or nonlinear-security analysis. The affine PLL injection also gives the consistency identity $\lambda_{\tau_i} = -\lambda(\lambda_{\log K_{p,i}} + \lambda_{\log K_{i,i}})$ at fixed ρ and trim. Multiple roots require subspace/nonsmooth analysis.

For $p = (\rho, \log K_p, \log K_i)$ a legitimate smooth local predictor is

$$\max_{\delta p} \sum_i P_i^0 \delta \rho_i, \quad g_j + \nabla_p g_j^T \delta p \leq 0, \quad p + \delta p \text{ inside gain bounds and a trust region.}$$

Spectral derivatives come from (45); nonlinear peak derivatives need the retarded variational equations and all active times. This is standard sequential constrained optimization, not a new global solver. No new optimized ρ, K_p, K_i are produced here.

Nor should $m_{\text{app}} \geq 0$ be added as a necessary safety constraint: overshoot below final frequency may still respect the allowed frequency excursion. Positive harmonic damping at sampled frequencies also does not prove stability.

G Exact retuning and its collective consequences

Scope. The design question is whether additional replacement can be compensated by bounded PLL gains, and which collective modes prevent that compensation. We keep the actual nonlinear DDAE, its exact delayed linearization and the physical ports. No new damping operator is postulated. All delays remain externally specified. A change of delay below compares two prescribed physical scenarios; it is not an optimization over latency. Every formula uses a fixed state-support chart.

G.1 The actual single-PLL return relation

Remove the three PLL states at site i and regard its angle θ_i as an input to the remaining grid. Keep all other PLLs, delays, SG, voltage, current and DC equations. At a regular frequency write

$$\mathcal{A}_i(s; \rho)z = \mathcal{B}_i(s; \rho)\theta_i, \quad e_i = \mathcal{C}_i(s; \rho)z + \mathcal{D}_i(s; \rho)\theta_i.$$

The exact return relation is

$$e_i = G_i(s; \rho)\theta_i, \quad G_i = \mathcal{D}_i + \mathcal{C}_i \mathcal{A}_i^{-1} \mathcal{B}_i. \quad (46)$$

The detector includes the negative local-angle term. G_i is not terminal voltage phase alone or a lossless scalar approximation. It is independent of this site's removed $K_{p,i}, K_{i,i}, \tau_i$; other controllers and delays are held fixed. Replacement dependence includes the actual algebraic closure and equilibrium dependence. Eliminating this PLL gives

$$F_i(s) = s^2(1 + t_{f,i}s) - (sK_{p,i} + K_{i,i})e^{-s\tau_i}G_i(s; \rho) = 0. \quad (47)$$

Where $\mathcal{A}_i$ is nonsingular, the full determinant is $\det(\mathcal{A}_i)F_i/t_{f,i}$ up to a nonzero ordering constant. Hidden-block singularities must instead be treated with the original characteristic operator. Not every pole need be visible in this relation.

Theorem 7 (Exact single-site pole assignment with gain limits). *Fix ρ , all other controllers and delays, and $\lambda = \alpha + i\omega$, $\omega > 0$. Assume $\mathcal{A}_i(\lambda)$ is invertible and $G_i(\lambda; \rho) \neq 0$. Define*

$$W_i(\lambda, \rho, \tau_i) = \frac{\lambda^2(1 + t_{f,i}\lambda)e^{\lambda\tau_i}}{G_i(\lambda; \rho)}.$$

The unique real gains assigning λ through this channel are

$$\boxed{K_{p,i} = \frac{\Im W_i}{\omega}, \quad K_{i,i} = \Re W_i - \frac{\alpha}{\omega} \Im W_i.} \quad (48)$$

Under declared gain bounds, such an assignment is possible if and only if

$$W_i \in \{\lambda k_p + k_i : k_p \in [K_p^-, K_p^+], k_i \in [K_i^-, K_i^+]\}. \quad (49)$$

Status: EXACT_IDENTITY.

Proof. Equation (47) reduces to $\lambda K_{p,i} + K_{i,i} = W_i$. Its imaginary and real parts give (48), uniquely since $\omega \neq 0$. The image of the gain rectangle is precisely the parallelogram in (49). $\square$

This is a one-pole, one-site feasibility test with other gains frozen. Failure is not a certificate against all distributed stabilization. Pole assignment is established control theory; the system-specific content is the exact return relation and the cross-mode result below.

G.2 Transport between prescribed heterogeneous delays

For frozen ρ and an existing root λ , compare τ_i with $\tau_i + h$. The exact gain map is

$$K_{p,i}(h) = e^{\alpha h} \left[K_{p,i} \cos \omega h + \frac{K_{i,i} + \alpha K_{p,i}}{\omega} \sin \omega h \right], \quad (50)$$

$$K_{i,i}(h) = e^{\alpha h} \left[K_{i,i} \cos \omega h - \frac{|\lambda|^2 K_{p,i} + \alpha K_{i,i}}{\omega} \sin \omega h \right]. \quad (51)$$

Indeed,

$$[\lambda K_{p,i}(h) + K_{i,i}(h)]e^{-\lambda(\tau_i+h)} = (\lambda K_{p,i} + K_{i,i})e^{-\lambda\tau_i}.$$

The eliminated PLL equation at λ is unchanged. Applying this map to all prescribed heterogeneous delay changes preserves the eliminated full-grid equations at $\lambda, \bar{\lambda}$, for fixed ρ . It does not ensure that these poles remain dominant or that the other poles remain stable. All gain bounds must still hold.

The same map solves

$$\frac{d}{dh} \begin{bmatrix} K_{p,i} \\ K_{i,i} \end{bmatrix} = \begin{bmatrix} 2\alpha & 1 \\ -|\lambda|^2 & 0 \end{bmatrix} \begin{bmatrix} K_{p,i} \\ K_{i,i} \end{bmatrix}. \quad (52)$$

At $\alpha = 0$, initially positive gains first lose positivity through $K_i = 0$ at $h = \omega^{-1} \arctan[K_i/(\omega K_p)]$ for increasing h . Actual bounds may be reached earlier. This is a boundary of the pole-preserving gain branch, not the maximum stable grid delay.

Theorem 8 (Modal spillover under single-site pole preservation). *Let $\lambda = \alpha + i\omega$ and μ be simple characteristic roots at the baseline, with return reductions regular at both. Retune site i by (51) as its prescribed latency changes, holding other sites fixed. Then*

$$\boxed{\left. \frac{d\mu}{dh} \right|_0 = -K_{p,i}(\mu - \lambda)(\mu - \bar{\lambda}) \frac{\partial \mu}{\partial K_{i,i}}}. \quad (53)$$

The derivative on the right is in ordinary gain, not log gain.

Status: EXACT_IDENTITY.

Proof. Differentiating (47) at a simple root gives

$$\mu_{K_{p,i}} = \mu \mu_{K_{i,i}}, \quad \mu_{\tau_i} = -\mu(K_{p,i} \mu_{K_{p,i}} + K_{i,i} \mu_{K_{i,i}}).$$

Combining these with (52),

$$\begin{aligned} \dot{\mu} &= \mu_{\tau_i} + (K_{i,i} + 2\alpha K_{p,i})\mu_{K_{p,i}} - |\lambda|^2 K_{p,i} \mu_{K_{i,i}} \\ &= -K_{p,i}(\mu^2 - 2\alpha\mu + |\lambda|^2)\mu_{K_{i,i}}. \end{aligned}$$

□

Equivalently, for $\mathcal{K}_i(s, h) = [sK_{p,i}(h) + K_{i,i}(h)]e^{-s(\tau_i+h)}$,

$$\partial_h \mathcal{K}_i(s, 0) = -K_{p,i}e^{-s\tau_i}(s - \lambda)(s - \bar{\lambda}).$$

For a finite change, $\mathcal{K}_i(s, h) - \mathcal{K}_i(s, 0)$ has zeros at $\lambda, \bar{\lambda}$, and is divisible by that quadratic as an entire function of s .

Corollary 9 (A precise single-site limitation). *If $\lambda_{K_{i,i}} \neq 0$, any differentiable real retuning at this site preserving the full complex root λ while $d\tau_i/dh = 1$ has derivative (52). For $K_{p,i} > 0$, it cannot also preserve a distinct root $\mu \notin \{\lambda, \bar{\lambda}\}$ to first order when $\mu_{K_{i,i}} \neq 0$.*

Proof. The gain-to-root real Jacobian is nonsingular: $\lambda_{K_i}(\lambda \delta K_p + \delta K_i)$ has nonzero real determinant for $\omega > 0$, $\lambda_{K_i} \neq 0$. Compensation is therefore unique; (53) is nonzero. $\square$

This is not an impossibility theorem for preserving only the real parts of two roots, or for distributed retuning at other sites. A complex root moving need not have a worsened real part. The sign of $\Re \dot{\mu}$ is a prediction to test, not an assumed destabilizing outcome.

Connection to Beyond Nodal Damping. The residue in (53) is

$$\mu_{K_{i,i}} = -\frac{\ell_\mu^* Z_{K_{i,i}}(\mu) r_\mu}{\ell_\mu^* Z_s(\mu) r_\mu}.$$

The local PLL fixes the spectral polynomial; the lossy network, inertia, P/Q and other devices fix its collective residue. Nodal damping coefficients do not contain these left/right overlaps. GSP can express the overlaps in a graph basis, but unsigned delay roughness cannot substitute for their complex sign.

G.3 Replacement at fixed physical delays

At any prescribed ρ in the regular region, (48) gives an exact candidate for one target pole. For fixed $\lambda, \tau_i, t_{f,i}$,

$$W_{i,\rho_j} = -W_i G_{i,\rho_j} / G_i, \quad K_{p,i,\rho_j} = \Im W_{i,\rho_j} / \omega, \quad K_{i,i,\rho_j} = \Re W_{i,\rho_j} - \alpha \Im W_{i,\rho_j} / \omega.$$

The derivative of G_i must include equilibrium and algebraic-voltage dependence. The sharing contract leaves $t_{f,i}$ fixed. Holding a full complex pole is one candidate strategy. For a requirement on real part alone, use $\lambda = -\sigma_{\text{req}} + i\omega$ with ω free; fixing ω would add an unrequested design constraint.

H From modal prediction to a bounded replacement decision

Theorem 10 (Gain-bounded first-order replacement in a fixed direction). *Start at a feasible design. Fix $\rho = \rho_0 + tr$, $t \geq 0$, with $P^T r > 0$, and an admissible interval $0 \leq t \leq t_{\text{box}}$. Stack the included smooth constraint branches as $g \leq 0$ and put $q = -g(p_0) \geq 0$, $a = g_\rho r$, $C = g_K$. Let the gain increments belong to a nonempty compact convex set $\mathcal{U}$ containing zero. In the first-order constraint model,*

$$\exists u \in \mathcal{U} : \quad ta + Cu \leq q$$

if and only if

$$t \eta^T a \leq \eta^T q + h_{\mathcal{U}}(-C^T \eta) \quad \text{for every } \eta \geq 0. \quad (54)$$

Consequently

$$t_{\max}^{\text{lin}} = \min \left\{ t_{\text{box}}, \inf_{\eta \geq 0, \eta^T a > 0} \frac{\eta^T q + h_{\mathcal{U}}(-C^T \eta)}{\eta^T a} \right\}. \quad (55)$$

An empty infimum is $+\infty$. The maximal extra replacement in this model is $t_{\max}^{\text{lin}} P^T r$ MW.

Status: PROVED_REDUCED_MODEL.

Proof. Feasibility means $q - ta \in \mathcal{CU} + \mathbb{R}_+^m$, a closed convex set. Its separating halfspaces with finite lower support use normals $\eta \geq 0$, and $\inf_{u \in \mathcal{U}} \eta^T Cu = -h_{\mathcal{U}}(-C^T \eta)$. This proves (54). For $\eta^T a \leq 0$ the inequality is automatic because $q \geq 0$, $0 \in \mathcal{U}$ and $t \geq 0$. The remaining inequalities give (55). $\square$

For the physical gain box $u_j \in [l_j, h_j]$, $h_{\mathcal{U}}(z) = \sum_j \max(l_j z_j, h_j z_j)$. The existing gain bounds are not enlarged. If $C^T \eta = 0$ for a nonnegative η , that weighted combination of constraints has no first-order tuning authority, even if gain increments were unbounded. Its replacement demand must be met by baseline margin or another physical intervention.

This is standard convex separation applied to the problem; it extends the repository's previous compensability formulation. It is not a new general optimization theorem, a maximum over all replacement directions, or a full nonlinear optimum. The physical contribution sought is how the collective residues determine the limiting combination of constraints. For spectral maxima or trajectory peaks, retain all relevant active branches or use directional nonsmooth analysis. A finite list does not certify unseen roots, time instants or disturbances.

For a uniformly verified remainder bound on the entire design region,

$$|g_j(p_0 + (tr, u)) - g_j(p_0) - ta_j - (Cu)_j| \leq \varepsilon_j,$$

replacing q by $q + \varepsilon$ gives a necessary relaxation, and hence a regional upper bound on t . Replacing q by $q - \varepsilon$ in (54) gives a sufficient test for the included constraints. The simplified ratio formula need not apply to $q - \varepsilon$, which can have negative entries. No rigorous remainder bound has been obtained here.

H.1 Analytic examples that expose the limiting mechanism

Let $t \geq 0$ denote normalized replacement and let $u \in [-\kappa, \kappa]$ be one normalized tuning direction. If the two modal constraints are

$$t + u \leq q, \quad t - u \leq q, \quad q \geq 0,$$

then adding them proves $t \leq q$, attained with $u = 0$. Each constraint separately would allow $t = q + \kappa$. The dual direction $\eta = (1, 1)$ cancels the controller action and detects the conflict. More tuning range does not remove this particular first-order multimode limitation. In contrast, if both rows are $t + u \leq q$, the optimum is $t = q + \kappa$, attained with $u = -\kappa$. These exact algebraic examples explain what the test detects; they do not assert that either row pattern occurs on IEEE-39.

H.2 A sufficient gate for the complete DDE spectrum

This standard homotopy argument prevents pole assignment from silently ignoring the other roots. At fixed ρ , in a common gauge-deflated $E = I$ state chart, let

$$\Delta_1(s) = \Delta_0(s) + U(s)C, \quad U_i(s) = b_i^0 e^{-s\tau_i^0} - b_i^1 e^{-s\tau_i^1}.$$

Both delay vectors are prescribed. Let Γ enclose $\Re s \geq -\sigma$, truncated by $|s| \leq R$, with Δ_0 nonsingular on Γ . A sufficient condition is

$$\sup_{s \in \Gamma} \|C\Delta_0(s)^{-1}U(s)\|_2 < 1. \quad (56)$$

Indeed, for $0 \leq t \leq 1$ the determinant lemma and Neumann invertibility make $\Delta_0 + tUC$ nonsingular on Γ . Its argument-principle root count is unchanged. There are no omitted roots for $|s| > R$, $\Re s \geq -\sigma$ if

$$R > \|A_0\|_2 + \max_{k=0,1} \sum_i \|b_i^k\|_2 \|c_i\|_2 e^{\sigma\tau_i^k}.$$

Dominance of sI proves this bound throughout the affine homotopy. A zero initial root count and verified (56) therefore imply the full spectral margin.

The gauge must actually be removed; pinned roots on the contour need separate deflation and are excluded here. This sufficient condition is potentially conservative and has not been evaluated. Sampling cannot certify its supremum. Changing ρ requires the full characteristic difference, including A_0, C and equilibrium dependence, rather than the fixed- ρ low-rank update above. Nonlinear event guards remain separate from this spectral argument.

Deferred validation; no experiments in this extension. First check the gain-sensitivity ratio and spillover formula on withheld simple modes. Then check finite gain transport, all gain bounds and the complete spectrum. Only afterwards test the frozen nonlinear events and whether a replacement predictor improves a feasible design under identical limits. No trajectories, optimization runs or spectral sweeps were executed here.

I All-PLL construction and its exact scope

Research contract. The primary question is whether the collective grid return can predict an admissible PLL retuning as SG replacement increases. The proposed mechanism is incompatibility between the gains demanded by different oscillatory patterns. The falsifying test is that this incompatibility fails to predict useful full-model designs, or disappears when unnecessary pattern restrictions are removed. The consequence, if validated, is a controller decision based on network interactions. Neighbor feedback is a separately identified architecture extension, with its own latency and gain contract. None of these statements assumes a strict advantage exists.

I.1 Open every PLL: the plant remains the full grid

Remove all PLL triples (θ, ω_p, ξ) while keeping the n angles as external inputs to the other device/network equations:

$$\mathcal{A}(s; \rho)z = \mathcal{B}(s; \rho)\theta, \quad e = \mathcal{C}(s; \rho)z + \mathcal{D}(s; \rho)\theta.$$

For $\mathcal{A}$ nonsingular, define

$$G(s; \rho) = \mathcal{D} + \mathcal{C}\mathcal{A}^{-1}\mathcal{B}, \quad D_f(s) = s^2(I + sT_f). \quad (57)$$

Unlike G_i in (46), G is independent of *all* removed PLL gains and detector delays. The current, DC, SG, lossy AC, P/Q and equilibrium dependencies remain in G . This independence uses the actual model contract: gains enter only the removed PLL equations. Do not silently reuse it for a model with other gain-dependent channels.

For local diagonal gains K_p, K_i , the exact reduced characteristic is

$$F(s; \rho, K) = D_f(s) - (sK_p + K_i)E_\tau(s)G(s; \rho), \quad E_\tau = \text{diag}(e^{-s\tau_i}). \quad (58)$$

Its determinant describes full-model roots wherever the eliminated block is regular; hidden-block singularities need the original descriptor. No approximation of the exponentials or modal diagonalization is used.

Theorem 11 (Analytical gains for a prescribed collective pattern). *Fix a replacement ρ , $\lambda = \alpha + i\omega$ with $\omega > 0$, and a nonzero PLL-angle vector $q \in \mathbb{C}^n$. Assume $\mathcal{A}(\lambda; \rho)$ is nonsingular and $y_i = [G(\lambda; \rho)q]_i \neq 0$ for every site. Set*

$$W_i = \frac{\lambda^2(1 + t_{f,i}\lambda)e^{\lambda\tau_i}q_i}{y_i}, \quad \boxed{K_{p,i} = \frac{\Im W_i}{\omega}, \quad K_{i,i} = \Re W_i - \alpha \frac{\Im W_i}{\omega}}. \quad (59)$$

These are the unique real local gains for which $F(\lambda)q = 0$. They are admissible precisely when they obey the declared gain bounds.

Status: EXACT_IDENTITY.

Proof. Row i of $F(\lambda)q = 0$ is $(\lambda K_{p,i} + K_{i,i})e^{-\lambda\tau_i}y_i = \lambda^2(1 + t_{f,i}\lambda)q_i$. Its two real equations give (59). Reconstruct $z = \mathcal{A}^{-1}\mathcal{B}q$, $\omega_p = \lambda q$ and $\xi = K_i E_\tau y / \lambda$ to recover the full eigenvector. Real model coefficients also give the conjugate eigenpair. $\square$

If $y_i = 0$ and the right side is nonzero, assignment is impossible. If both vanish, that row imposes no gain equation and must be treated without division. Nonzero q is required; its complex scale cancels in (59). This result assigns an eigenpair, not dominance, simple multiplicity, the complete spectrum, or nonlinear safety. Conversely, any existing local-gain design with such an eigenpair and nonzero responses satisfies this same formula. Thus allowing the pole and pattern to vary covers this regular eigenpair chart; holding a convenient pattern fixed is the additional restriction.

The collective dependence is explicit:

$$y_i = \sum_j G_{ij}(\lambda; \rho) q_j.$$

Thus even local gains depend on the spatial amplitudes and phases of the whole pattern. Replacing this sum by $G_{ii}q_i$ is an additional model approximation. Offline use of G does not require online communication or make the controller distributed.

I.2 Exact gain compatibility for several patterns

For m distinct nonreal target pairs (λ_r, q_r) , construct W_{ir} from (59). Let

$$C_\Lambda = \begin{bmatrix} \alpha_1 & 1 \\ \omega_1 & 0 \\ \vdots & \vdots \\ \alpha_m & 1 \\ \omega_m & 0 \end{bmatrix}, \quad w_i = \begin{bmatrix} \Re W_{i1} \\ \Im W_{i1} \\ \vdots \\ \Re W_{im} \\ \Im W_{im} \end{bmatrix}.$$

Then, with no gain bounds, simultaneous local assignment is possible if and only if

$$\boxed{(I - C_\Lambda C_\Lambda^\dagger)w_i = 0 \quad \text{for all } i.} \quad (60)$$

The gains are $[K_{p,i}, K_{i,i}]^T = C_\Lambda^\dagger w_i$. Since $\omega_r > 0$, C_Λ has rank two and the gains are unique; bounded feasibility also requires membership in the gain rectangle.

Status: EXACT_IDENTITY. The proof is the range criterion for $C_\Lambda k_i = w_i$. For two pairs this is four real equations for two real gains at each site. The conflict is therefore testable, rather than inferred from large or negative damping coefficients.

Scope restriction. Failure proves infeasibility for these *specified poles and patterns*. It does not prove that local controllers cannot stabilize the same replacement with other modes. It is not a decentralized fixed-mode theorem. Patterns and target frequencies must be allowed to vary before claiming an architectural stabilization limit.

I.3 Which additional measurement can remove a conflict?

For an admissible directed information pattern $\mathcal{N}_i$, including the local signal, consider the explicit extension

$$\begin{aligned} t_{f,i}\dot{\omega}_{p,i} &= \xi_i - \omega_{p,i} + \sum_{j \in \mathcal{N}_i} k_{p,ij} e_j(t - \tau_j - d_{ij}), \\ \dot{\xi}_i &= \sum_{j \in \mathcal{N}_i} k_{i,ij} e_j(t - \tau_j - d_{ij}), \quad \dot{\theta}_i = \omega_{p,i}. \end{aligned}$$

Every $d_{ij} \geq 0$ is an exogenous communication delay, with $d_{ii} = 0$. Signals are calibrated detector errors with declared units, sign and timestamps. Their zero locked values preserve the equilibrium. Off-diagonal gains and their bounds define a new controller contract; they must not be treated as the old twenty-gain architecture.

For fixed target (λ_r, q_r) define

$$y_r = G(\lambda_r; \rho) q_r, \quad z_{ijr} = e^{-\lambda_r(\tau_j + d_{ij})} (y_r)_j, \quad b_{ir} = \lambda_r^2 (1 + t_{f,i} \lambda_r) (q_r)_i.$$

The row condition is

$$\sum_{j \in \mathcal{N}_i} [\lambda_r z_{ijr} k_{p,ij} + z_{ijr} k_{i,ij}] = b_{ir}.$$

Stack real and imaginary parts into the real linear system

$$A_i x_i + B_i u_i = b_i. \tag{61}$$

A_i contains the two self-gain columns; B_i contains the proposed neighbor-gain columns. A proportional column for j is the stacking of $\Re(\lambda_r z_{ijr})$, $\Im(\lambda_r z_{ijr})$; an integral column stacks $\Re z_{ijr}$, $\Im z_{ijr}$. Apply a fixed declared row scaling before using residual norms or singular values. Exact feasibility is unchanged by invertible row scaling.

Theorem 12 (Compatibility and minimum added gain). *For (61), let $\Pi_i = I - A_i A_i^\dagger$, $r_i = \Pi_i b_i$ and $H_i = \Pi_i B_i$. Assume initially that self gains are unrestricted. There exist real gains realizing every prescribed pair at site i if and only if $r_i \in \text{range } H_i$. With added-gain metric $R_i \succ 0$, the unique minimum-metric neighbor gain is*

$$u_i^* = R_i^{-1} H_i^T (H_i R_i^{-1} H_i^T)^\dagger r_i, \tag{62}$$

$$\kappa_{i,\min}^2 = r_i^T (H_i R_i^{-1} H_i^T)^\dagger r_i. \tag{63}$$

Self gains then solve $A_i x_i = b_i - B_i u_i^*$.

Status: EXACT_IDENTITY.

Proof. Project (61) onto the orthogonal complement of $\text{range } A_i$. This gives $H_i u_i = r_i$, which is also sufficient because the remaining residual belongs to $\text{range } A_i$. Whiten u_i with $R_i^{1/2}$ and apply the minimum-norm solution of a consistent linear system; this proves both formulas. Its nullspace is orthogonal in the metric to the displayed solution, proving minimality and uniqueness. $\square$

An added budget $u_i^T R_i u_i \leq \kappa_i^2$ is feasible if and only if the range test holds and $\kappa_{i,\min} \leq \kappa_i$, still with unrestricted self gains. With bounds on either gain family, solve the exact convex feasibility/QP problem (61) with those bounds; the displayed unconstrained minimizer is not sufficient. Polynomial graph coefficients shared across sites require stacking all rows into one linear problem, rather than independent row solves.

With $d_i = |\mathcal{N}_i|$ incoming signals, the assignment matrix has $2d_i$ columns and $2m$ rows. Arbitrary modal demands require $d_i \geq m$ and full row rank. One neighbor can resolve a two-pair conflict only when its two columns supply the missing directions; communication delays can destroy or improve this rank/conditioning. A graph edge count alone is insufficient. These statements concern prescribed eigenstructure, not global stabilizability.

I.4 Why arbitrary powers of a Laplacian do not close the problem

Let L be a normalized, symmetric, connected Laplacian and $K(L)$ a polynomial graph filter. For delayed source measurements,

$$[L, K(L)E_\tau(s)] = K(L)[L, E_\tau(s)]. \quad (64)$$

If $K(L)$ is invertible, the left side vanishes exactly when the rightmost commutator vanishes. Its entry is $L_{ij}(e^{-s\tau_j} - e^{-s\tau_i})$. Commutation throughout a frequency interval therefore forces equal delays along all edges, hence uniform delay on a connected graph. At isolated frequencies, phase coincidence can be an exception. Furthermore,

$$\|[L, E_\tau(i\Omega)]\|_F^2 = 8 \sum_{\{i,j\} \in \mathcal{E}} w_{ij}^2 \sin^2\left(\frac{\Omega(\tau_i - \tau_j)}{2}\right). \quad (65)$$

Status: EXACT.IDENTITY. Both formulas follow entrywise; each undirected edge occurs twice. If $\sigma_{\min}(K(L)) \geq \gamma > 0$, the remaining commutator norm is at least $\gamma\|[L, E_\tau]\|_F$. This is a channel-mixing result, not a stability lower bound. It does not apply unchanged to nonuniform diagonal gains plus graph terms, or to general edge-delayed filters.

The physical network basis U is useful for expressing $q = Ua$ and decomposing Gq . Using all graph coordinates is an exact basis change; truncating a restricts the design and needs separate evidence. The actual model need not commute with L , even at zero delay. Powers L^ℓ encode weighted walks, not an independent physical cycle basis. No new instability is established by such powers alone.

I.5 A calculable replacement predictor and corrector

Use (59) as an exact gain-elimination map $K = \Phi(\rho, \lambda, q)$ on a regular chart. Normalize one nonzero complex component of q to one. For fixed delays and nonzero q_i, y_i, W_i , its differential is

$$dW_i = W_i \left[\left(\frac{2}{\lambda} + \frac{t_{f,i}}{1 + t_{f,i}\lambda} + \tau_i \right) d\lambda + \frac{dq_i}{q_i} - \frac{dy_i}{y_i} \right], \quad (66)$$

$$dy = G dq + G_s q d\lambda + \sum_j G_{\rho_j} q d\rho_j, \quad (67)$$

$$dK_{p,i} = \frac{\Im(dW_i) - K_{p,i} d\omega}{\omega}, \quad (68)$$

$$dK_{i,i} = \Re(dW_i) - \alpha dK_{p,i} - K_{p,i} d\alpha. \quad (69)$$

Use the undivided quotient rule at a zero numerator. Derivatives of G include the algebraic closure and any equilibrium dependence. The chain rule for another simple root is

$$d\mu = \sum_j \mu_{\rho_j} d\rho_j + \sum_i (\mu_{K_{p,i}} dK_{p,i} + \mu_{K_{i,i}} dK_{i,i}), \quad \mu_p = -\frac{\ell^* Z_p r}{\ell^* Z_s r},$$

where the physical-port representation must be regular; otherwise use the full descriptor sensitivity. This retains the collective, reactive and delay effects, including the spillover in (53).

In normalized real coordinates $\zeta = (\rho, \alpha, \omega, \Re q_{\text{free}}, \Im q_{\text{free}})$, let $p(\zeta) = (\rho, \Phi(\rho, \lambda, q))$. The replacement problem on this chart is

$$\min_{\zeta} \sum_i P_i^0(1 - \rho_i), \quad g_j(p(\zeta)) \leq 0, \quad \alpha \leq -\sigma_{\text{req}},$$

with the actual gain/replacement bounds, every required physical root, and all nonlinear event guards included in g . This is an exact reparameterization on the declared chart, not a convexification of the security constraints. A computable local subproblem is

$$\max_{\delta\zeta} (P^0)^T \delta\rho - \frac{\beta}{2} \delta\zeta^T R_{\zeta} \delta\zeta, \quad g_k + J_{g,k} J_{p,k} \delta\zeta \leq 0, \quad \delta\zeta \in \mathcal{T}_k. \quad (70)$$

Here $\beta > 0$, $R_{\zeta} \succ 0$ and the convex trust region $\mathcal{T}_k$ are declared in advance. It is a convex QP when $\mathcal{T}_k$ is polyhedral. Include all active smooth branches of peak/spectral constraints; a single gradient of their maximum is not generally valid. J_p uses (69). This is a standard SQP predictor with an analytical gain map.

The method to implement after the algebra gate is:

1. Start from a fully feasible design and a simple visible oscillatory root, with its actual PLL pattern. Confirm reconstruction by (59); this parity test is mandatory.
2. At a frozen ρ -direction, compute the predicted gain change from (69). Alternatively allow ρ to vary in a trust region and maximize $(P^0)^T d\rho$ using all active linearized security constraints after substituting dK .
3. Permit ω , the admissible real part, and the complex pattern to vary. Fixing them is a declared restricted predictor, not a fair upper bound on local-controller capability.
4. Evaluate a finite trial through $K = \Phi(\rho, \lambda, q)$, rather than applying only the tangent gain update. This makes the chosen eigenpair exact within arithmetic/model error. Reject bound violations and singular or poorly conditioned charts.
5. Check the complete required spectral region and the frozen nonlinear guards. If another mode becomes limiting, include it and use the compatibility equations with variable patterns in a local Newton/SQP correction. Reduce the step when predicted and actual constraint changes disagree.
6. Stop at a declared local stationarity/residual criterion, a chart failure, or an unresolved feasibility conflict. Record which. Do not report a restricted-family stop as the full replacement maximum.

For prescribed patterns, the gain solve is explicit or convex. Jointly varying patterns and replacement makes the outer problem nonconvex. With all q coordinates free, this parameterization does not promise dimension reduction or computational superiority. Local Newton convergence requires a nonsingular normalized residual Jacobian; root collisions require a subspace/nonsmooth method. No global convergence or quantitative speedup is asserted.

No frequency, RoCoF, voltage or actuator guard is inferred merely from pole assignment. Their original nonlinear definitions and all frozen events remain mandatory at the later validation stage. A sufficient full-spectrum continuation condition was given above; it has not been evaluated. Exact local gains alone are not an exact security frontier.

I.6 Evidence gates and literature boundary

Pole assignment with delay, eigenstructure design, sparse feedback and spectral optimization are established methods [8, 9, 10, 11]. Ram et al. treat delayed feedback in symmetric mechanical systems; Katewa and Pasqualetti treat sparse static state-feedback assignment; Vanbiervliet et al. treat nonsmooth spectral optimization for retarded systems, including continuation of nonlinear equilibria. These antecedents preclude claiming that the matrix solve, projection, or iterative optimization is itself new. The narrower candidate is the bounded PLL compatibility law under the full-grid return and its verified consequence for replacement or latency decisions. Recent converter-model comparisons reinforce the need to retain hidden dynamics [12]; supplementary GFL power-oscillation controllers [13] are a different action set from PLL-only tuning.

Before new grid simulations, close three gates: (i) the exact gain map, compatibility and derivatives pass algebra checks; (ii) the controller architecture, fixed delays, bounds, target-selection rule and falsification criteria are written down; (iii) the first validation has a numerical prediction to test, not a desired replacement percentage. Gate (i) is mathematical only; IEEE-39 parity, all-root checks and event validation were the subsequent gates. Section 6 reports their targeted execution.

The preregistered first experiment tests whether the formula reconstructs known gains and predicts a finite replacement correction on withheld modes. Only after that succeeds should a matched-budget design comparison run. Neighbor feedback is investigated only if the local modal conflict is material after relaxing arbitrary pattern choices. A rigorous claim that neighbors enable otherwise impossible secure replacement would need a valid local-architecture upper bound below a fully feasible distributed design; fixed-pattern incompatibility is not that bound.

References

- [1] R. Pates and E. Mallada, *Damping, Inertia, and Delay Robustness Trade-offs in Power Systems*, MTNS, 2018. <https://mallada.ece.jhu.edu/pubs/2018-MTNS-PM.pdf>.
- [2] F. Paganini and E. Mallada, *Global performance metrics for synchronization of heterogeneously rated power systems: The role of machine models and inertia*, arXiv:1710.07195v2, 2017. <https://arxiv.org/html/1710.07195v2>.
- [3] L. Pagnier and P. Jacquod, *Optimal placement of inertia and primary control: a matrix perturbation theory approach*, arXiv:1906.06922, 2019. <https://arxiv.org/abs/1906.06922>.
- [4] L. Huang, H. Xin, W. Dong, and F. Dörfler, *Impacts of Grid Structure on PLL-Synchronization Stability of Converter-Integrated Power Systems*, arXiv:1903.05489, 2019. <https://arxiv.org/abs/1903.05489>.
- [5] J. Wilson, A. Nelson, and B. Farhang-Boroujeny, *Parameter Derivation of Type-2 Discrete-Time Phase-Locked Loops Containing Feedback Delays*, IEEE TCAS-II, 56(12):886–890, 2009. <https://doi.org/10.1109/TCSII.2009.2034197>.
- [6] P. Zítek, J. Fišer, and T. Vyhlídal, *Dimensional analysis approach to dominant three-pole placement in delayed PID control loops*, Journal of Process Control, 23(8):1063–1074, 2013. <https://www.sciencedirect.com/science/article/abs/pii/S0959152413001236>.
- [7] W. Zhang, Y. Cui, and X. Ding, *An Improved Analytical Tuning Rule of a Robust PID Controller for Integrating Systems with Time Delay Based on the Multiple Dominant Pole-Placement Method*, Symmetry, 12(9):1449, 2020. <https://doi.org/10.3390/sym12091449>.

- [8] Y. M. Ram, J. E. Mottershead, and M. G. Tehrani, *Partial pole placement with time delay in structures using the receptance and the system matrices*, Linear Algebra and Its Applications, 434(7):1689–1696, 2011. <https://doi.org/10.1016/j.laa.2010.07.014>.
- [9] V. Katewa and F. Pasqualetti, *Minimum-Gain Pole Placement With Sparse Static Feedback*, IEEE Transactions on Automatic Control, 66(8):3445–3459, 2021. <https://doi.org/10.1109/TAC.2020.3018615>.
- [10] J. Vanbiervliet, K. Verheyden, W. Michiels, and S. Vandewalle, *A nonsmooth optimisation approach for the stabilisation of time-delay systems*, ESAIM: COCV, 14(3):478–493, 2008. <https://doi.org/10.1051/cocv:2007060>.
- [11] X. Mao and H. Dai, *Partial eigenvalue assignment with time delay robustness*, Numerical Algebra, Control and Optimization, 3(2):207–221, 2013. <https://doi.org/10.3934/naco.2013.3.207>.
- [12] S. Coffey, S. Harrison, A. Egea-Alvarez, and L. Xu, *Assessing the Capability and Limitations of Small Signal Simplified Grid Following Models for the Identification of Sub-Synchronous Oscillations*, IEEE Access, 2025. <https://doi.org/10.1109/ACCESS.2025.3586197>.
- [13] T. Chakraborty, A. Pal, and S. Maleki, *Damping LFOs: Grid Following with Power Oscillation Damping vs. Grid Forming vs. PSS*, arXiv:2505.24204, 2025. <https://arxiv.org/abs/2505.24204>.
- [14] W.-J. Beyn, *An integral method for solving nonlinear eigenvalue problems*, arXiv:1003.1580, 2010. <https://arxiv.org/abs/1003.1580>.
- [15] J.-P. Lessard and J. D. Mireles James, *A functional analytic approach to validated numerics for eigenvalues of delay equations*, Journal of Computational Dynamics, 7(1):123–158, 2020. <https://doi.org/10.3934/jcd.2020005>.
- [16] W. Michiels and S. Gumussoy, *Eigenvalue based algorithms and software for the design of fixed-order stabilizing controllers for interconnected systems with time-delays*, in Delay Systems: Advances in Delays and Dynamics, Springer, 2014; author version arXiv:2003.05496. https://doi.org/10.1007/978-3-319-01695-5_18.
- [17] U. Marković, O. Stanojev, P. Aristidou, E. Vrettos, D. Callaway, and G. Hug, *Understanding Small-Signal Stability of Low-Inertia Systems*, IEEE Transactions on Power Systems, 36(5):3997–4017, 2021. <https://doi.org/10.1109/TPWRS.2021.3061434>.
- [18] Z. Chu and F. Teng, *Stability Constrained Optimization in High IBR-Penetrated Power Systems—Part I: Constraint Development and Unification*, author version, arXiv:2307.12151. <https://arxiv.org/abs/2307.12151>.
- [19] H. Jia et al., *Converter-Driven Stability Constrained Generation Expansion for High IBG-Penetrated Power Systems*, IEEE Transactions on Power Systems, 2025. <https://doi.org/10.1109/TPWRS.2025.3558879>.
- [20] D. Peaucelle and Y. Ebihara, *Robust stability analysis of discrete-time systems with parametric and switching uncertainties*, IFAC World Congress, 2014. <https://skoge.folk.ntnu.no/prost/proceedings/ifac2014/media/files/0282.pdf>.
- [21] D. Bindel and A. Hood, *Localization Theorems for Nonlinear Eigenvalue Problems*, SIAM Journal on Matrix Analysis and Applications, 2013. <https://www.cs.cornell.edu/~bindel/papers/2013-simax.pdf>.

- [22] F. Borgioli, W. Michiels, D. Lu, and B. Vandereycken, *A globally convergent method to compute the real stability radius for time-delay systems*, author manuscript, 2019. https://www.unige.ch/math/vandereycken/papers/published_Borgioli_MLV.pdf.
- [23] F. Johansson, *Arb: Efficient Arbitrary-Precision Midpoint-Radius Interval Arithmetic*, IEEE Transactions on Computers, 66(8):1281–1292, 2017. <https://doi.org/10.1109/TC.2017.2690633>.